

Heavy Flavour Hadronisation and Balancing Studies as a Function of Multiplicity in pp Collisions at the LHC with PYTHIA 8.3

MSc Thesis

written by

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Auf Weiss II, Kandinsky

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Abstract

Hadronisation describes the formation of mesons and baryons from coloured partons. It is fundamental in understanding how matter is created directly after the Big Bang. It is described by Quantum Chromodynamics (QCD), which is non-perturbative at low energies. Unfortunately for us, hadronisation takes place in this non-perturbative region and therefore the study is limited to phenomenological models in Monte Carlo (MC) event generators.

Previous studies have shown that these MC models have difficulty in predicting the creation of charm baryons. However, the Junctions hadronisation model of the PYTHIA MC generator shows a promising baryon increase and I extend these studies to beauty and charm quarks, including multiplicity dependence. I confirm the expected baryon increase in Junctions and I comment on differences in beauty and charm. An interesting peak-like behaviour is seen for the baryon/meson ratios as a function of multiplicity, which is not expected naively.

My results mainly probe PYTHIA, though when the future ALICE upgrade is realised, these simulation studies can also be compared to data.

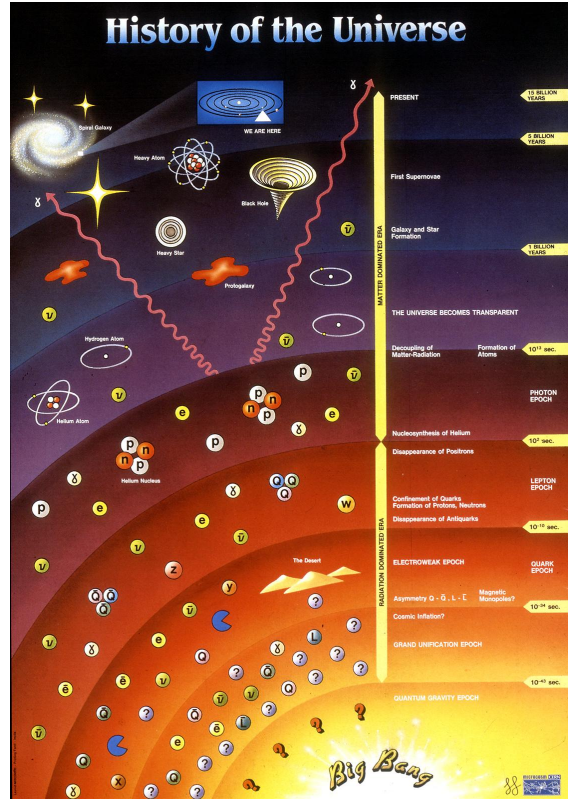
Contents

1	Introduction	1
2	Theoretical Background	3
2.1	PYTHIA Event Generator	9
3	Analysis	13
3.1	Experimental Observables	13
3.2	Analysis Details	19
4	Results	23
4.1	Angular correlations of B^+B^- and D^+D^- meson pairs	23
4.2	Balancing of beauty and charm hadrons	26
4.3	Baryon/meson balancing ratio of beauty and charm hadrons	27
5	Conclusion and Outlook	30
A	PYTHIA simulation settings	35
B	D^+D^- angular correlations of mothers	37

1 Introduction

It has long been established that the Universe and everything around us was created in a single event about 13.7 billion years ago, the Big Bang [1–3]. The circumstances under which the Big Bang took place were among the most extreme that can be imagined with our current understanding of physics. The extreme temperatures and density during this event are expected to have formed a new state of matter in which even the atomic nuclei themselves melt into their constituents, the *Quark Gluon Plasma* (QGP) [4]. The QGP eventually *froze-out* due to decreasing temperature caused by the expansion of the Universe, forcing the quarks to form bound states in a process called *hadronisation* [5].

The study of hadronisation is one of the central physics topics of the ALICE (A Large Ion Collider Experiment) experiment [7] at CERN, Geneva. The ALICE experiment tries to understand the strong nuclear force and the emergent phenomena that took place in the earliest seconds of the Universe. In order to mimic the situation that existed right after the Big Bang in an experiment, it is necessary to attain a sufficiently high energy that is capable of breaking apart the colliding particles. The ALICE experiment realises this by making use of the Large Hadron Collider (LHC) at CERN, the world’s most energetic particle collider, capable of reaching center-of-mass energies of up to 13.6 TeV for proton-proton (pp) collisions. The LHC has allowed ALICE to study the strong force in even more sophisticated detail with respect to previous accelerators, pushing our understandings to new heights [8].



Timeline of the development stages from the Big Bang until the present day. The transition from quark epoch to the lepton epoch after about 10^{-10} seconds signals the occurrence of the hadronisation process. Image courtesy of CERN [6].

The hadronisation mechanism is governed by the strong force, described in the Standard Model of particle physics as a gauge theory called *Quantum Chromodynamics* (QCD). A special characteristic of QCD is that its interaction strength is not constant, but rather dependent on the scale of the energy transfer in the interaction. The interaction strength gets weaker at higher energy scales, unlike the more familiar electromagnetic interaction, which is a phenomenon called *asymptotic freedom* [9, 10]. At lower energies, however, the interaction strength becomes stronger and below a certain scale the perturbative expansion series become divergent and QCD loses its predictive power. This non-perturbative region requires the application of phenomenological machinery using Monte Carlo (MC) event generators and lattice QCD. As hadronisation binds quarks together at scales much greater than the typical partonic distances, it falls in the regions of non-perturbative QCD. It is due to the non-perturbative side of QCD that the fundamental nature of all matter around us, even the creation and stability of the most primitive particles - the protons - are not understood.

Heavy-flavour (beauty, charm) quarks are used in this analysis to study hadronisation in pp collisions at the LHC. The simulations are done with PYTHIA 8.3 [11] and are agnostic to the current ALICE experiment as the study focuses on the MC level. The study is motivated, however, by the ALICE3 upgrade [12] which will provide the statistics and coverage to reconstruct the necessary amount of heavy-flavour baryons to be able to compare them with phenomenological predictions.

A further reason to use heavy quarks to study hadronisation is that their mass forces them to be produced in the initial high energy scattering processes of the collision, such that their origin can be pinned down cleanly. Moreover, the parton production is described by perturbative QCD and can be calculated up to excellent precision [13, 14], while the production and hadronisation calculations can be split and considered independently in an approximation called *factorisation* [15].

The goal of this thesis is to explore how hadronisation is modelled with different hadronisation mechanisms as a function of the charged-particle event multiplicity. The study of the multiplicity dependence is interesting as it relates strongly to the multiple parton interactions (MPIs) that form an integral part of hadronic collisions and are related to the hadronisation of quarks into baryons. The creation of baryons with event generators is challenging to predict in comparison to mesons [16]. The Monash [17] and Junctions [18] tunes of PYTHIA are studied in in order to address the baryon deficit that is seen in simulations when they are compared to data.

Another reason to study the multiplicity dependence relates to recent signs of possible collective behaviour effects in smaller collision systems with large event multiplicities [19]. This discussion is out of scope for this analysis, but it forms a natural continuation to this study.

The structure of this thesis is as follows. Sec. 2 expands the theoretical topics that have been introduced in this section, in particular regarding QCD and PYTHIA. Afterwards, the analysis structure and method is given in Sec. 3. The results are then shown and discussed in Sec. 4 and the thesis is completed with a conclusion and outlook in Sec. 5.

2 Theoretical Background

Quantum Chromodynamics (QCD) is the theoretical framework describing the strong interaction between quarks and gluons. The central idea is that the quarks and gluons have another quantum number, *colour*, which is denoted as red (R), green (G) or blue (B) and that the dynamics of these coloured particles can be described by a gauge theory based on the SU(3) (special unitary) symmetry group. The 8 gauge bosons of this symmetry group are interpreted as the gluons and their colours are not just R, G or B, but they are combinations of colour and anti-colour to match the adjoint representation of the group. An important property of colour is that coloured states can never be isolated, called *colour confinement*, and instead states must consist either of the same colour and anti-colour, or a combination of all three different colours, called *white* [20].

The discussion in this section is based to a large extent on Refs. [20–26]. References are also indicated in the text for specific paragraphs or statements in order to illustrate clearly which sources were used in each part.

Similar to the other gauge theories in the Standard Model, the theory can be written down straightforwardly from the SU(3) group, but this results in diverging physical observables. In order to re-express the divergences of the theory in a well-defined way, it is required to implement a thorough *renormalisation* procedure [25]. One way to achieve this renormalisability is to introduce a new arbitrary energy scale μ that manages to absorb regularised divergences into physical quantities. One of those physical quantities that depends on this scale is the strong coupling constant. Any renormalisation procedure with any μ must result in the same physics, from which it is possible to write down the evolution or *running* of the coupling constant α_s as a function of the renormalisation scale μ . Assuming three colours (Standard Model), the evolution is described by

$$\alpha_s(Q^2) \equiv \frac{g^2(Q^2)}{4\pi} = \frac{12\pi}{(33 - 2n_f) \ln(Q^2/\Lambda^2)}, \quad (1)$$

where Q^2 is the energy transfer of the process, n_f is the number of quark flavours, Λ is the energy scale and $g(0)$ is the value of the coupling constant at a reference scale, which comes from experiment. The running of the strong coupling in QCD is illustrated in Fig. 3.

It is evident that α_s becomes smaller for increasing Q^2 and goes to zero in the limit $Q^2 \rightarrow \infty$, which is called *asymptotic freedom* [9, 10]. However, when the energy transfer is smaller than the renormalisation scale of QCD, $\Lambda \equiv \Lambda_{QCD} \approx 200 \text{ MeV}$ [23], α_s diverges and perturbation theory breaks down. This distinctively splits QCD into two regions: the non-perturbative region for $Q \leq \Lambda_{QCD}$ and the perturbative region where $Q > \Lambda_{QCD}$ [20, 23, 25].

Hadronisation is described by the lower-energy regime and is hence governed by non-perturbative QCD, restricting its investigation to the study of phenomenological models. One of the main arguments to study hadronisation with heavy-flavour quarks is that due to their mass they can be described by the higher-energy regime and are thus controlled by perturbative QCD. Assuming the production and hadronisation processes are independent, called the *factorisation* approach [15], the analysis simplifies considerably.

Fig. 4 gives an intuitive visualisation for charm hadrons into how the factorisation processes are related and separated. The particle collision consists of a hard interaction between two partons that take a momentum fraction of the colliding protons. Their behaviour is described with parton distribution functions (PDFs). The partons themselves collide in the perturbative regime of QCD, which is able to calculate the interaction cross-sections of these partons. The coloured partons that

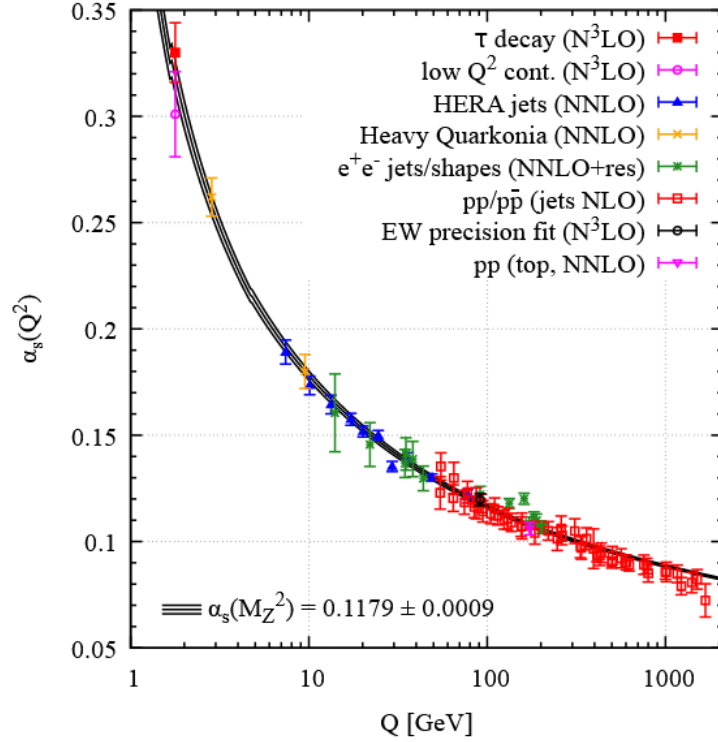


Figure 3: Summary of measurements of the strong coupling constant α_s as a function of the energy scale Q . The relative degree of perturbation theory is shown in brackets (NLO: next-to-leading order, NNLO: next-to-next-to-leading order, N³LO: next-to-NNLO, NNLO+res: NNLO matched to a resummed calculation). It is evident that the coupling diverges at low energies and perturbation theory starts to break down [27].

are created in the hard-scattering and other processes during the collision will start hadronising and bind together to form colour singlet final states. Each of these steps is assumed to be independent in the factorisation approach and this makes the calculation of the total cross-section manageable. The low and high energy regions of the process will now be addressed in a bit more detail.

Following the chronology of the pp collision, the discussion starts in the perturbative region. Heavy-flavour quarks are only produced in hard-scattering processes, specifically in the initial hard-scattering of the collision, due to their large masses: charm ~ 1 GeV, beauty ~ 4 GeV and top ~ 170 GeV [27], though the top quark is too heavy and will decay through the weak interaction before it is able to hadronise [26]. The energy transfer in these processes is well above the QCD scale Λ_{QCD} and perturbative methods can be applied. Typical hard-scattering processes [21] that produce heavy quarks (Q) in hadron-hadron collisions are the quark-antiquark and gluon-gluon pair production, but there exist also higher-order contributions from flavour excitation and gluon splitting processes, illustrated in Figs. 5a - 5d.

The relative abundances of the cross-sections of the typical heavy-quark production processes can be calculated, though this subdivision is unobservable and dependent on the model parameters

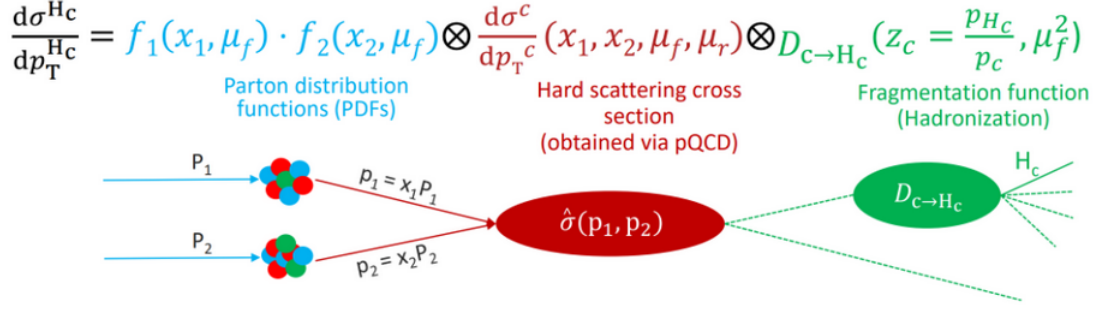


Figure 4: Sketch of the factorisation principle in the calculation of the differential cross-section of a given charm hadron species (H_c). The parton distribution functions (PDFs) are controlled by the momenta fractions x_1 , x_2 of the interacting partons and kept at a certain renormalisation scale μ_f . Perturbative calculations are possible in the hard-scatterings of the partons, whereas the non-perturbative hadronisation step requires additional experimental input on the probability of a charm quark to fragment into H_c . The steps are each assumed independent in this assumption so that the cross-section becomes a product of the individual steps.

used. It is still illustrative to show these results to indicate consistency within the PYTHIA model and the agreement with theory. The calculations were done in Ref. [28] for beauty and charm cross-sections in pp collisions as a function of their center-of-mass energy $\sqrt{s}$ and the results are given in Fig. 6. The results align well with data, which is seen by comparison with Fig. 7 and therefore it sounds compelling that all significant contributions to the heavy-flavour quark production originate from the production processes mentioned. Regarding the behaviour of the cross-sections, for very small energies, the pair production mechanism is dominating the total production cross-section, while flavour excitation contributions are substantial and gluon splitting is negligible. The trend reverses at the highest energies, where gluon splitting becomes the largest contribution, followed by flavour excitation and pair production. This behaviour is intuitive, as any hard-scattering process will contain one of the leading-order $2 \rightarrow 2$ scatterings at any energy, while the number of branches in the initial and final state showers will increase with energy. Flavour excitation topologies are more restrictive than gluon spitting mechanisms, although their kinematic requirements are lower at more limited energies, explaining the observed difference. The differences in beauty and charm are due to their mass difference and the beauty case essentially gets shifted to higher energies with respect to charm. This limits the statistics that can be achieved for the beauty correlations by a factor of 10^4 ($= (10^2)^2$) as compared to charm. Finally, it is also interesting to compare the beauty and charm production processes and their cross-sections at LHC energies. Around the pp LHC energy of 14 TeV, the pair creation processes are taken over by gluon splitting (for charm) and flavour excitation (for beauty) becoming subdominant for the latter, whereas flavour excitation was already dominant over pair production for charm.

Continuing the chronology of a typical pp collision, the hard-scattering is followed by the non-perturbative fragmentation processes to collect the partons and form colourless bound states. The description of this hadronisation procedure is dependent on models, although the form of the potential that binds quarks together is motivated by studies done in lattice QCD [32–34] and seems to be constantly increasing as a function of distance. This behaviour suggests that the

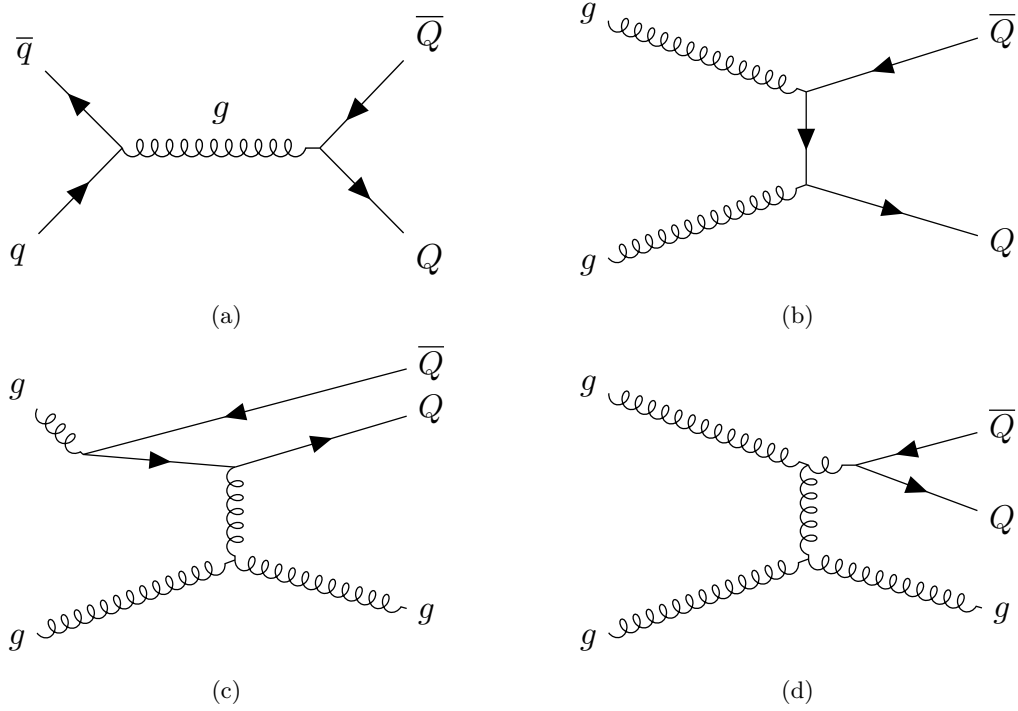


Figure 5: Examples of diagrams in the production of heavy-flavour quarks in hadronic collisions ($Q = c, b$). Dominant (a) - (b) pair production diagrams and next-to-leading order (c) flavour excitation and (d) gluon splitting diagrams are shown [28].

hadronisation potential can be modelled in a string-like way, which led to the development of the Lund string model [35].

In the Lund model, illustrated in Figs. 8 and 9, the potential between two quarks is modelled as a string and the creation of new quark-antiquark pairs is realised by the breaking of these strings, which occurs when the energy in the colour field between the pair is sufficiently large [35]. This string breaking probability occurs through the tunnelling of a potential barrier which is proportional to $\exp(-\pi m^2/0.2(\text{GeV}/c)^2)$. The probability to form new pairs scales strongly with the mass m squared of the particle pair involved and hence disfavours the production of heavy quark pairs. It can be derived [35] that the relative abundances for the first quark pairs are of the order: $uu : dd : ss : cc \sim 1 : 1 : 1/3 : 10^{-11}$ and even more extreme for bb pairs. Clearly, virtually no heavy-flavour quark pairs are produced through the string breaking mechanism and rather they are produced in the hard-scattering processes in the earliest stages of the event. This provides a clear signature for the origin of the heavy quark.

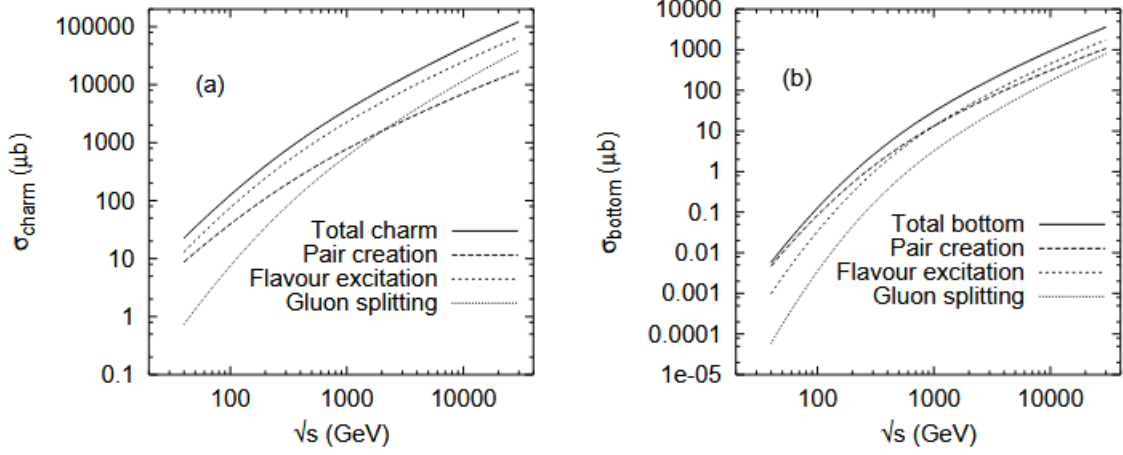


Figure 6: The total (a) charm and (b) beauty production cross-sections for pp collisions as a function of center-of-mass energy $\sqrt{s}$. The contributions from pair creation, flavour excitation and gluon splitting are shown separately [28].

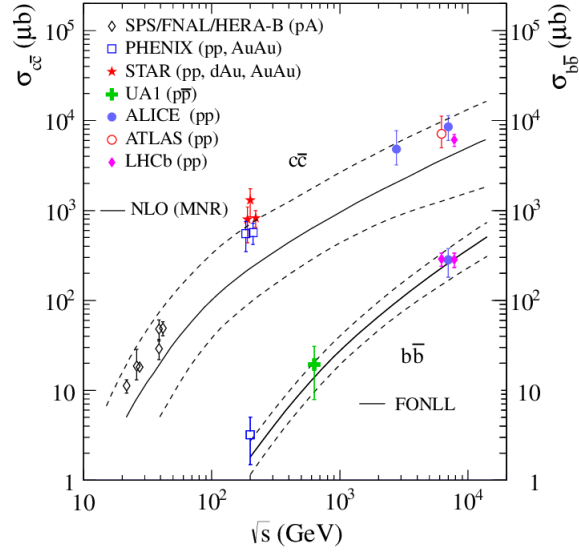


Figure 7: The total charm and bottom production cross-sections for pp collisions as a function of center-of-mass energy $\sqrt{s}$. Experimental data is shown and compared with the NLO (MNR) [29] and FONLL [30] calculations. The errors are depicted by dashed lines. References to all mentioned collaboration results can be found within the paper [31].

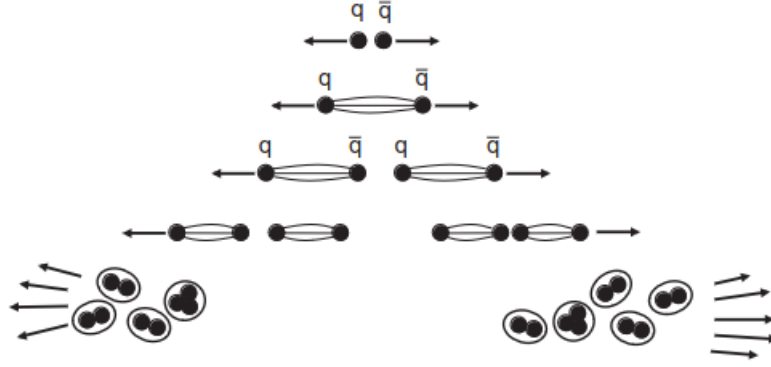


Figure 8: In the Lund String Model, string breaks produce new quark-antiquark pairs from the energy that is stored in the colour field connecting the quarks and subsequently fragment into hadrons [26].

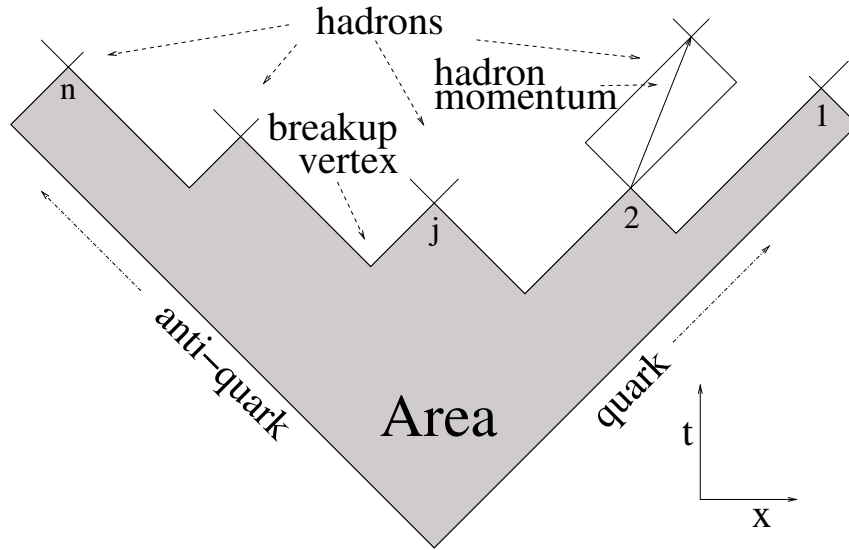


Figure 9: Depiction illustrating the chronological and spatial progression of consecutive string breaks. Quarks originating from neighbouring breaking points combine to create hadrons [36].

2.1 PYTHIA Event Generator

In order to describe the non-perturbative QCD process of hadronisation, Monte Carlo event generators can be applied to simulate and study particle collisions and the subsequent formation of quarks into hadrons. PYTHIA 8.3 [11] is used in this analysis for the simulation of pp collisions and creation of colourless beauty and charm hadrons. The general flow of the event simulation procedure will be given in this section with a particular focus on the models implemented to describe the hadronisation processes, as described in Refs. [11, 17, 18]. In order to increase the clarity and convenience of reading in the text, PYTHIA 8.3 will be denoted by PYTHIA onwards.

The event generator process in PYTHIA can be divided into three main parts: process level, parton level, and hadron level, sketched in Fig. 10. The **process level** represents the hard-scattering process, which is described perturbatively. The **parton level** includes the evolution and dynamics at parton level. Initial- and final state radiation, multiparton interactions (MPIs), beam remnants and colour-reconnection are treated. At the end of this stage, the event exhibits a realistic parton structure. Finally, the **hadron level** is concerned with the hadronisation phenomena to confine

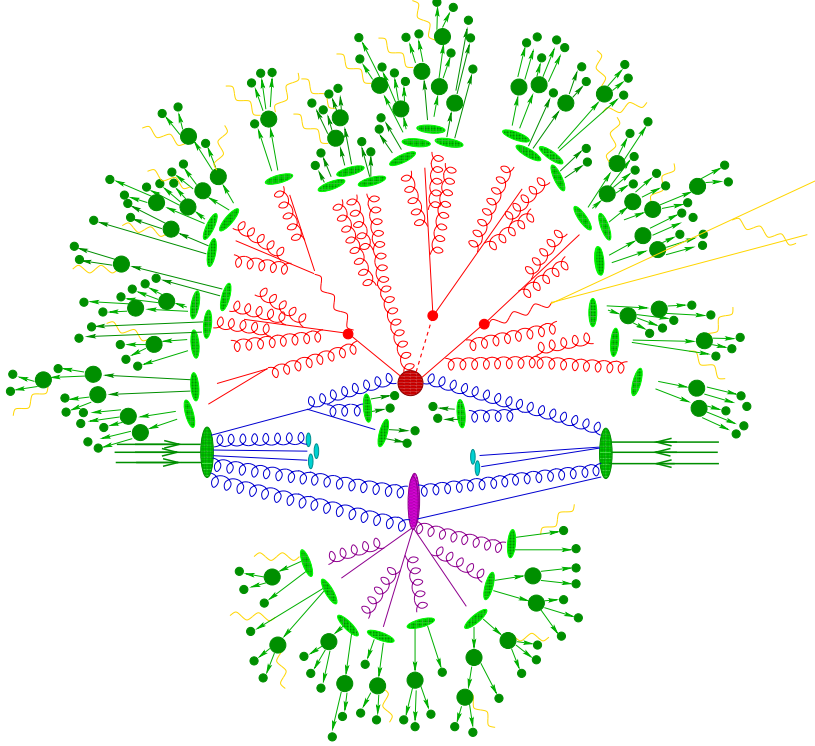


Figure 10: Illustration of a particle collision and the different event sub-processes in PYTHIA. The red part represents the hard parton interaction and evolution into showers. The purple part represents a second partonic interaction, which increases the event multiplicity. The hadronisation processes (light-green) and final state hadrons and associated decays (dark-green) are shown and conclude the event simulation process [37].

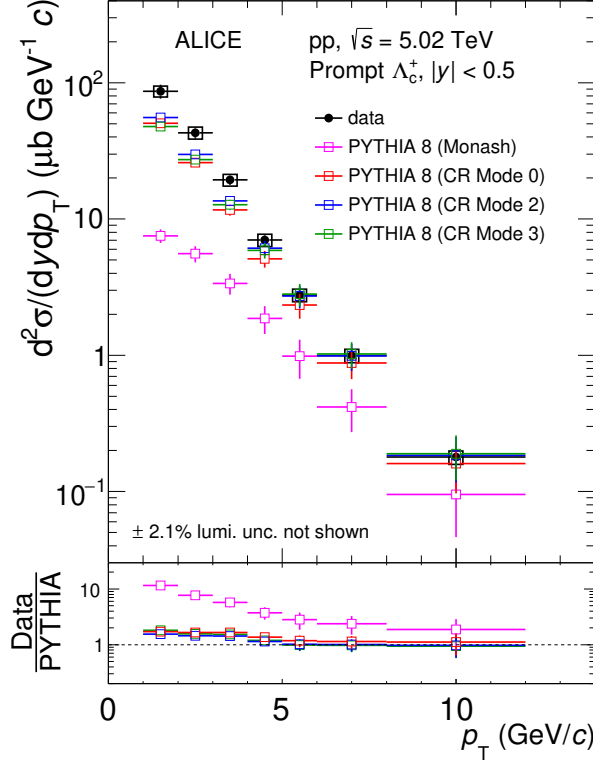


Figure 11: An ALICE measurement [16] of the prompt Λ_c^+ -baryon p_T -differential production cross-section in pp collisions at $\sqrt{s} = 5.02$ TeV is shown. The statistical uncertainties are shown as vertical bars and the systematic uncertainties are shown as boxes. Comparison to predictions from different tunes of the PYTHIA 8 event generator are done [17,18]. The vertical bars on the PYTHIA 8 predictions represent the statistical uncertainty from the simulation, and the vertical bars on the ratios in the bottom panel also include the statistical uncertainties from the data.

the partons into hadrons, which is described non-perturbatively with the Lund model [11].

PYTHIA performs adequately in studies of mesons but shows difficulties in similar studies with baryons, in particular under-estimating their production yield. This under-estimation is consistent with what can be expected from the Lund model, as strings can only break to form quark-anti-quark pairs, which means that three of these pairs are necessary to form a baryon and an anti-baryon, while mesons with only one pair. This trend has been observed in experiments [16] and can be seen clearly in Fig. 11 where the production yield of the Λ_c^+ baryon is compared for different extensions of the PYTHIA 8.3 generator.

Effort has been made, however, to increase the production of baryons in PYTHIA. One way to increase these production rates is to *tune* parameters in the event generator. Many of these parameters are inaccessible from theoretical considerations and must be tuned using data from experiments. The collection of tuned parameter is then called a *tune*. A particularly strong tune was

developed for PYTHIA 8.1 [17] with the purpose of increasing baryon production, called *Colour Reconnection* (CR), and hints for CR processes were found already in [38]. CR methods attempt to aid the hadronisation process by allowing the merging of partons originating from separate MPI systems and not only from the hard-scattering. This was developed with e^+e^- in mind and the generator output agrees excellently with data [17]. This hadronisation method is still an approximation, however, as the limit $N_C \rightarrow \infty$ is taken. This limit corresponds with connecting each coloured quark to one unique other coloured quark in the partonic final state. In general, beyond leading colour (LC) effects are suppressed by the number of colours squared $1/N_C^2$, because colour-connected partons from different MPIs need to be connected.

On the other hand, in hadronic collisions the situation is different from the e^+e^- system. First, the initial state and beam remnants now include coloured partons, which need to be taken into account. Moreover, high multiplicity events are dominated by contributions from MPIs and effects from different MPI systems must be considered in the hadronisation procedure. This is motivated by hints from hadron collision data that collective hadronisation effects are at play [18, 38–41]. One of those hints is the dependence of the average charged particle $\langle p_T \rangle$ on the event multiplicity. Measurements of this quantity show that $\langle p_T \rangle$ increases linearly as a function of multiplicity, which implies that there are no independently hadronising MPI systems, for which this dependence would be flat.

Beyond leading colour CR effects are necessary to account for the hadronisation mechanisms in hadronic collisions. Multiple extensions were proposed [18] but the focus of this thesis is on the most promising, *CR mode 2*, referred to as *Junctions* throughout the text. The original PYTHIA 8.1 version will be referred to as Monash and is used as baseline for comparison. Within Junctions, hadronisation is not only possible indirectly through string breaking and merging from separate MPI systems, but rather also directly through the formation of junction-type topologies, increasing the baryon production yield. Just prior to hadronisation, the leading-colour strings are transformed into new configurations based on three principles: the SU(3) colour rules from QCD that decide whether two strings are compatible, the requirement of causal contact between the strings and finally, the λ measure, a string-length measure, is used to check if a configuration is actually energetically favoured [42, 43]. Four different types of reconnections can occur, shown in Fig. 12. Note that although predominantly $q\bar{q}$ strings are drawn, all combinations (including qq , $q\bar{q}$, $\bar{q}q$ and gg) are treated identically. A **Type I** dipole reconnection in which two dipoles exchange endpoints, with a probability of $P_I^{q\bar{q}} = 1/9$ for $q\bar{q}$, $P_I^{qg} = 1/3$ for qq and $P_I^{gg} = 1/8$ for gg . This has been compromised in the model into a probability of $P_I = 1/9$ for all dipole-type reconnections. **Type II** and **Type III** can form junction anti-junction structures from two or three dipoles, and similarly, this has been compromised into $P_{II} = 2/9$ and $P_{III} = 2/81$ probabilities, respectively. The remaining case, the **Type IV** reconnection, allows two string systems to form junction anti-junctions at different points connected by their gluons. The probability P_{IV} for this to occur depends on the number of gluons. The probabilities denote only the chances of these recombination structures being created and it can be seen that the chance to create junction anti-junction connections is larger compared to the ordinary dipole connections. It must be noted however, that the string minimisation λ is applied after this recombination and typically disfavours the junction-type connections unless a strong overlapping of phase space is present, as in the case of the creation of baryons or with significant contribution of MPIs. [18].

The main reason the hadronisation study is done as a function of multiplicity is related to the MPIs and their role in the junction topologies. Hadrons are composite objects, in the case of protons in the LHC consisting of three valence quarks, sea quarks and gluons. When two protons

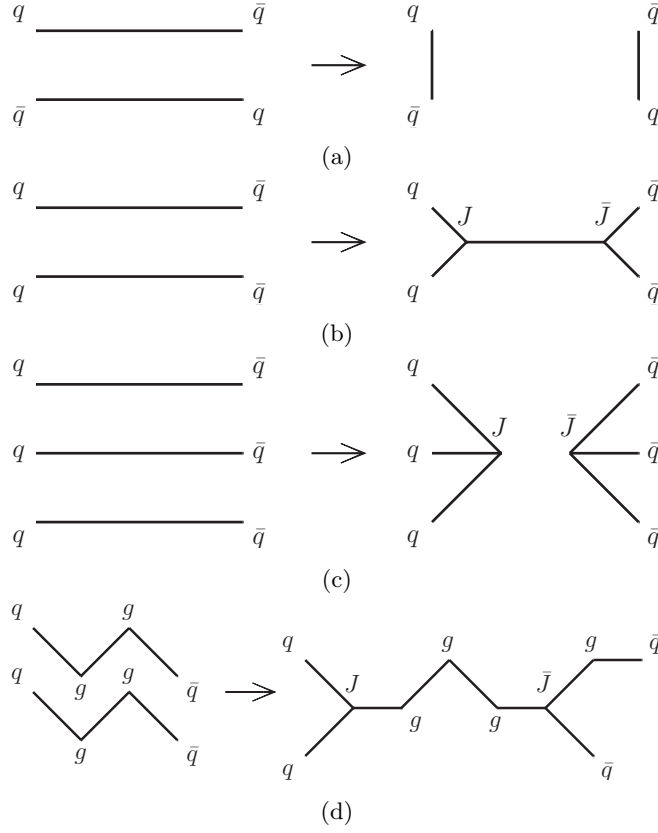


Figure 12: The four different allowed reconnection types. Type I (a) is the ordinary string reconnection. Type II (b) is the formation of a connected junction anti-junction pair. Type III (c) is the formation of junction and anti-junction, which are not directly connected. Type IV (d) is similar to type II except that it allows for gluons to be added between the two junctions. Figure and caption from [18].

collide, there are plenty of opportunities for MPIs, resulting in larger multiplicities. Events with larger multiplicities are therefore expected to show an increased amount of baryons compared to events with lower multiplicities.

3 Analysis

3.1 Experimental Observables

The observables that are studied in this analysis are the two-particle correlations in azimuthal space and the charged-particle event multiplicity. The angular correlations are used to calculate balancing yields of a given trigger particle (B^+ or D^+ meson) for several associate particles and their baryon-over-meson ratios. These variables are defined in this section.

All angular correlations consist of a trigger and associate particle that form a pair. Each trigger is selected in the simulation when it is identified as the required trigger and satisfies certain phase cuts (Sec. 3.2) motivated by the future upgrade [12] of the ALICE detector. Associate particles are then matched with the given trigger and satisfy the same cuts. Following this selection, the azimuthal angular difference of the trigger and associate pair, denoted $\Delta\phi$, is calculated and plotted in the range of $-\pi/2$ to $3\pi/2$ in order to visualise the peaks around $\Delta\phi = 0$ and $\Delta\phi = \pi$:

$$\Delta\phi = (\phi_1 - \phi_2) \quad (2)$$

The normalised correlation spectra are obtained by dividing the number of pairs by the number of trigger particles. If there are additional conditions on the trigger and associate (e.g. within a certain multiplicity range) then the normalisation is performed over the number of triggers in that range. The normalised per-trigger yield Y is therefore given by:

$$dY = \frac{1}{N_{\text{triggers}}} \frac{d^2N}{d\Delta\eta d\Delta\phi}. \quad (3)$$

An example of a two-particle angular correlation spectrum is given in Fig. 13. The away-side peak around $\Delta\phi = \pi$ is expected as heavy-flavour particles are exclusively produced in the initial hard-scattering processes of the pp collision (energy conservation) where the outgoing momenta must be equal with opposite sign (momentum conservation), hence their production topology is predominantly back-to-back. The emerging near-side peak at $\Delta\phi = 0$ for charm mesons is an artefact of the non-prompt contribution to the inclusive charm production and is discussed in Sec. 4.1. All correlations are projected in the angular plane; the pseudorapidity dependence is not taken into account in this study in order to focus on the effects of the different simulation tunes and multiplicity.

The information obtained with the angular correlation spectra can be used to study the *balancing* of a given trigger particle by different associates. Each trigger originates from the quark-antiquark pair that is created in the initial stage of the event. An associate particle is coupled to this trigger and a correlated pair is formed. A like-sign (LS) pair is identified as a hadron pair where both hadrons have the same heavy-flavour quark content, while unlike-sign (US) pairs are defined for opposite (quark-antiquark) content. The physical correlations are hence from US pairs, while LS pairs belong to a part of the background of the event. Examples of LS and US correlations are given in Fig 15. The LS correlations are easily identified, showing essentially flat distributions, and can be removed to reduce background.

The total background consists of random correlations between particles of a given type not associated with any common particle production mechanism. The method of removing LS pairs typically does not cover all background and the amount which is removed is strongly dependent on the `PhaseSpace:pTHatMin` parameter in PYTHIA, discussed in Sec. 3.2.

The cleanliness of the heavy-flavour production origin is used to study the hadronisation product

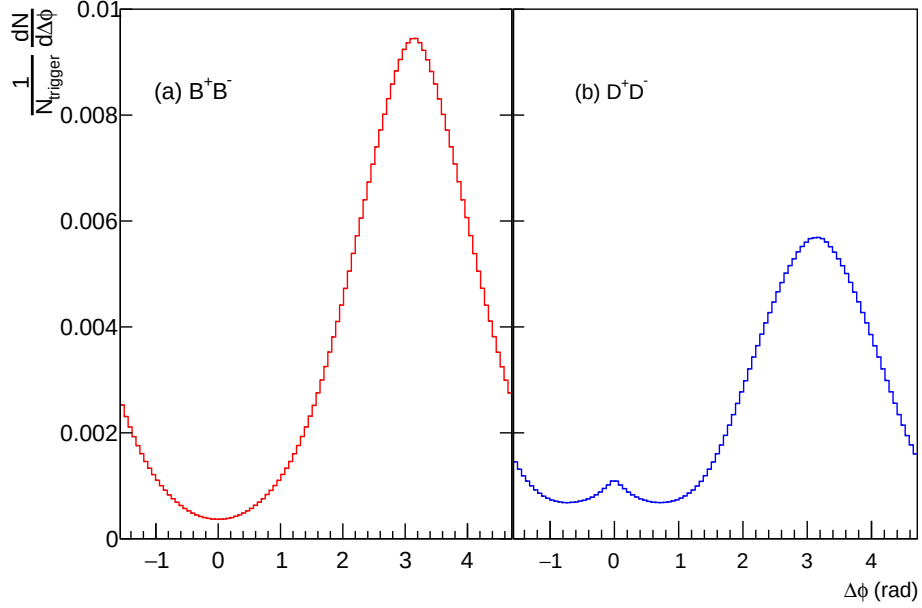


Figure 13: Angular correlations for (a) B^+B^- and (b) inclusive D^+D^- mesons over the 2π azimuthal range as simulated by PYTHIA with the Monash tune. The histograms are normalised by the total number of trigger B^+ or D^+ mesons, respectively. Distinct away-side peaks are observed at $\Delta\phi = \pi$ and a hint of a near-side peak is seen for charm mesons at $\Delta\phi = 0$

of the other quark in this original pair, interpreted as the balancing partner of the trigger particle. The balancing yield is obtained by integrating the background-suppressed angular correlation spectrum over the whole $\Delta\phi$ range, though selected integration ranges can also be considered. The pseudorapidity is again projected out.

Explicitly:

$$Y = \int_0^{2\pi} dY = \frac{1}{N_{\text{triggers}}} \int_0^{2\pi} d\Delta\eta d\Delta\phi \left[\frac{d^2N}{d\Delta\eta d\Delta\phi} \right]. \quad (4)$$

An example of balancing is given in Fig. 14. The stark differences in yield for the meson and baryon associates are caused by the differences in mass between the particles and the difficulty of producing baryons compared to mesons in Monash. On the other hand, the differences between beauty and charm are minor. Balancing is the main topic of this thesis and will be discussed in great detail in Sec. 4.2 and 4.3.

Experimental searches benefit from ratios of observables for several reasons. The main benefit is the cancellation of systematical errors and the experimental availability of observables. Related to this, common quantities that are unknown or hard to calculate in the individual observables

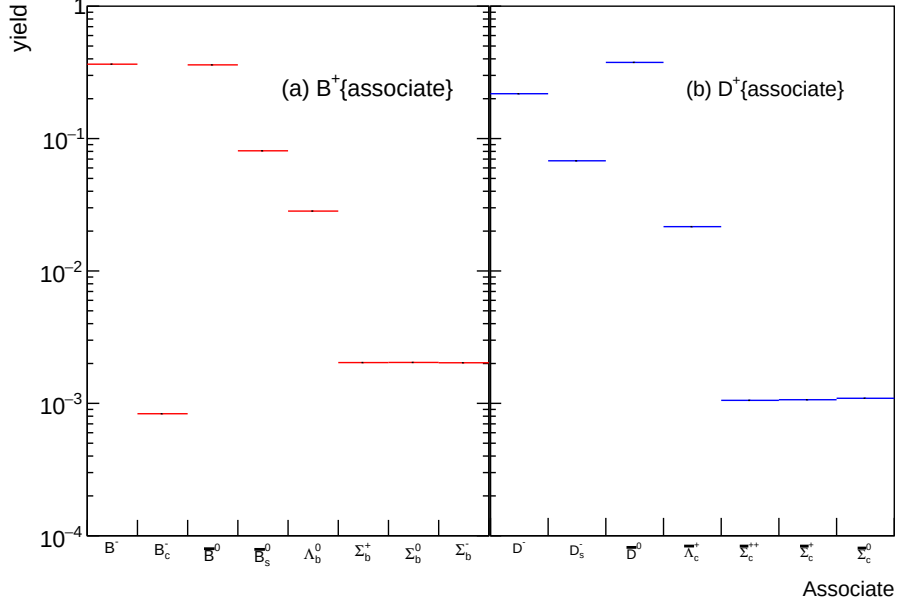


Figure 14: Full azimuthal-integrated angular correlation yields show balancing of (a) B^+B^- and (b) inclusive D^+D^- mesons, as simulated by PYTHIA with the Monash tune. A clear mass effect is seen in the balancing, as well as a preference for meson partners.

will also cancel out. An example of this is the production cross-section of beauty quarks, which will be divided out in the ratio of e.g. $B^+\Lambda_b^0$ over B^+B^- pairs, including any uncertainties that are associated with these variables. Another motivation is to aid the reader in understanding the physical meaning of results shown and to shed light on otherwise obscured trends in a figure. The balancing ratio of baryons-over-mesons relies on the last two arguments, as statistics nor experimental availability form issues for a simulation-driven study.

The motivation to study hadronisation in PYTHIA comes from an under-estimation of baryon production in simulations with Monash, which is expected to increase with Junctions and as a function of increasing event multiplicity. Therefore, it is interesting to compare the balancing yield of the trigger h with a baryon b divided by the balancing yield with a meson m , i.e.

$$Y_{b/m}(h) = \frac{Y(\{h\}\{b\})}{Y(\{h\}\{m\})}. \quad (5)$$

The baryon/meson ratio observable is to be interpreted as the performance of the given tune at producing more baryons to balance a given trigger. This comparison is especially relevant with Junctions compared to Monash. These results are discussed in Sec. 4.3.

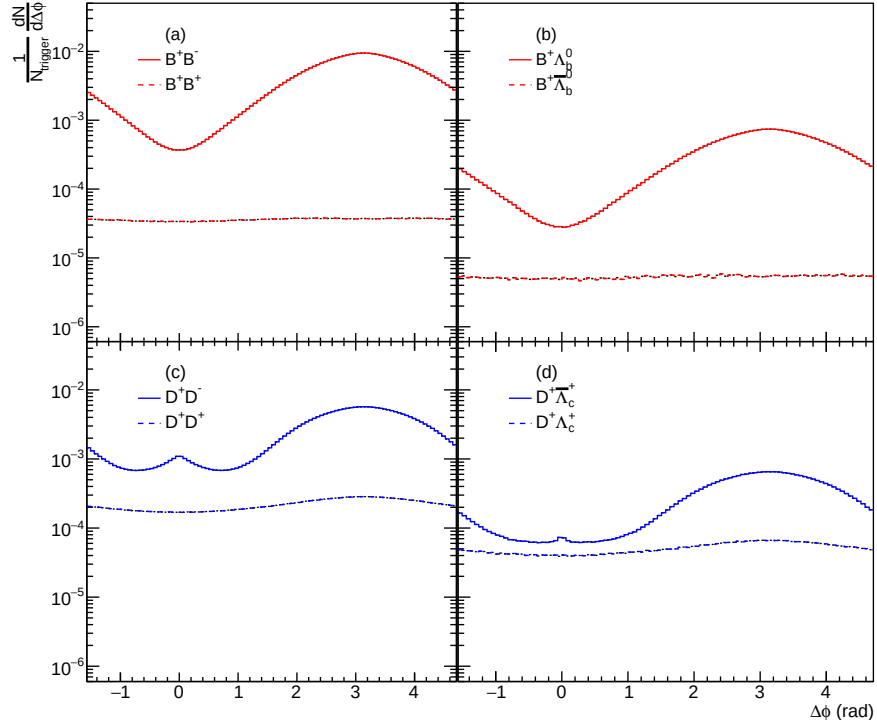


Figure 15: Angular correlations for like-sign (LS, dashed) and unlike-sign (US, solid) (a) B^+B^- , (b) $B^+\Lambda_b^0$, (c) inclusive D^+D^- and (d) inclusive $D^+\bar{\Lambda}_c^+$ (lower) mesons over the 2π azimuthal range as simulated by PYTHIA with the Monash tune. The histograms are normalised by the total number of trigger B^+ or D^+ mesons, respectively. The LS correlations are background and are characterised by flat distributions.

All observables are presented as a function of charged-particle event multiplicity, considered a proxy of MPIs in PYTHIA. It is expected that events with higher multiplicities had more overlap in phase space at the partonic stage and hence would produce more junction topologies, increasing the number of baryons created in the collision. Multiplicity can be defined in several ways. The definition which is used in this analysis is that multiplicity is defined as the number of primary charged particles passing a 0.15 GeV p_T and $|\eta| < 4$ (pseudorapidity) cut, where primaries are π , p , K , e , or μ particles originating from the initial scattering of the event or from strong decays. Multiplicity is collected for all events and afterwards a binning is set up that categorises events in their relative abundances in a way to be compatible with the definition of centrality in heavy-ion collisions. The 1% of events with the highest multiplicity are stored in the top 1% multiplicity bin, while the one with the 20% lowest multiplicity are stored in the 80-100% bin. The multiplicity spectra for beauty and charm quarks, including the used multiplicity intervals, are given in

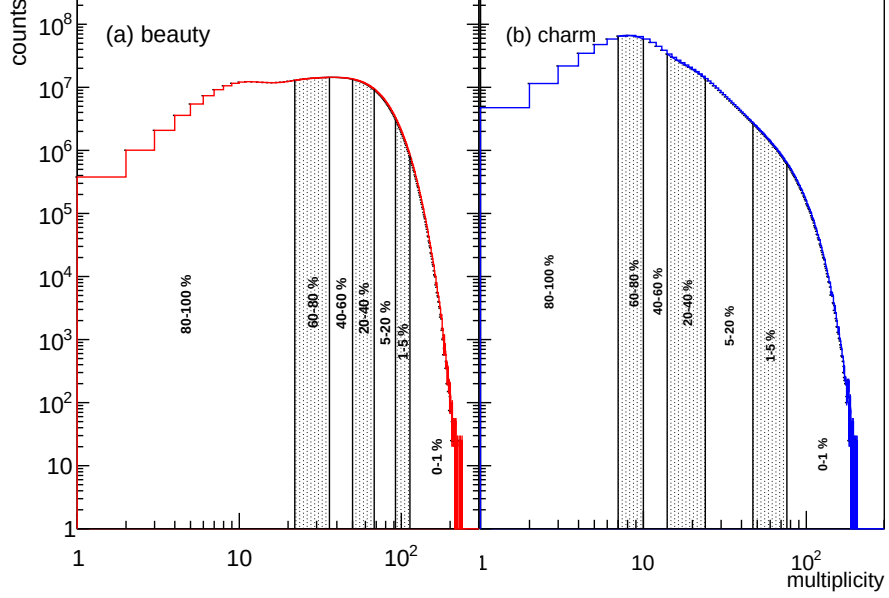


Figure 16: Multiplicity spectra of beauty and **prompt** charm quarks as produced with the `HardQCD:hardbbbbar` and `HardQCD:hardccbar` settings in PYTHIA Monash. The indicated multiplicity intervals are used in this analysis.

Fig. 16. The dependency of the angular correlations on multiplicity is illustrated in Fig. 17. Note the differences in beauty and charm, the increase in yield as a function of multiplicity, and the pronounced near-side peak for the highest multiplicities in D^+D^- meson correlations. The shapes of the multiplicity spectra are related due to the fact that biased productions are used to enhance statistics. Comparisons in beauty and charm within the defined intervals are not necessarily fair in that regard, though the qualitative behaviour remains the same.

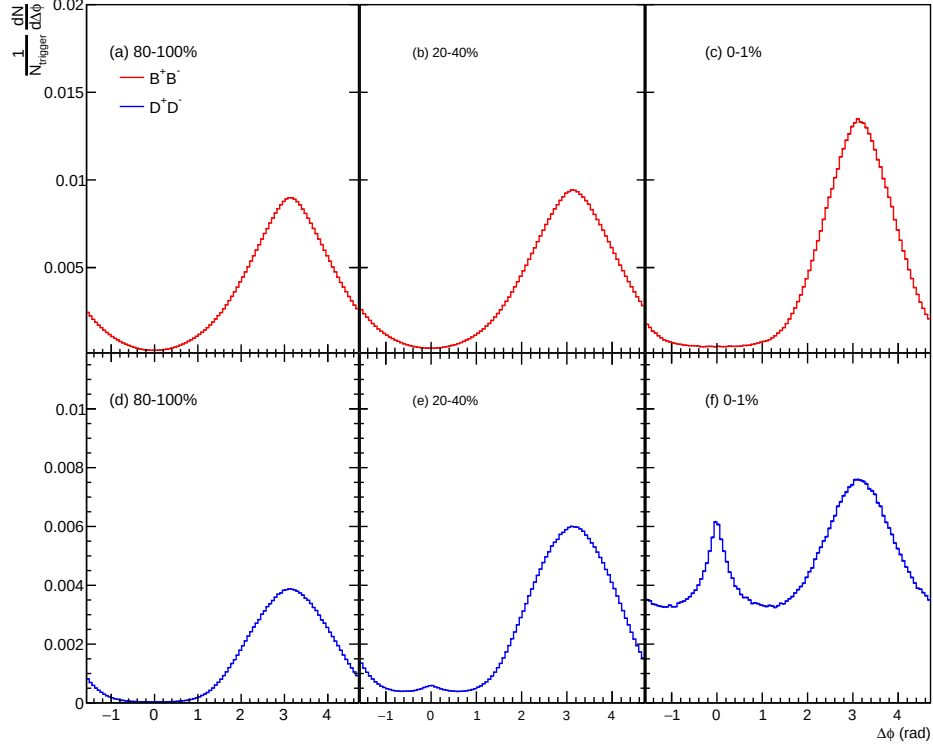


Figure 17: Angular correlations for (a) - (c) B^+B^- and (d) - (f) inclusive D^+D^- mesons over the 2π azimuthal range as a function of charged-particle event multiplicity, simulated by PYTHIA with the Monash tune. The multiplicity binning is described in the text, but can be interpreted as increasing from left to right. The histograms are normalised by the total number of trigger B^+ or D^+ mesons, respectively. An increase in per-trigger yield is observed for beauty and charm correlations, while the D^+D^- meson correlations show a distinctly different behaviour in the highest multiplicity bin.

3.2 Analysis Details

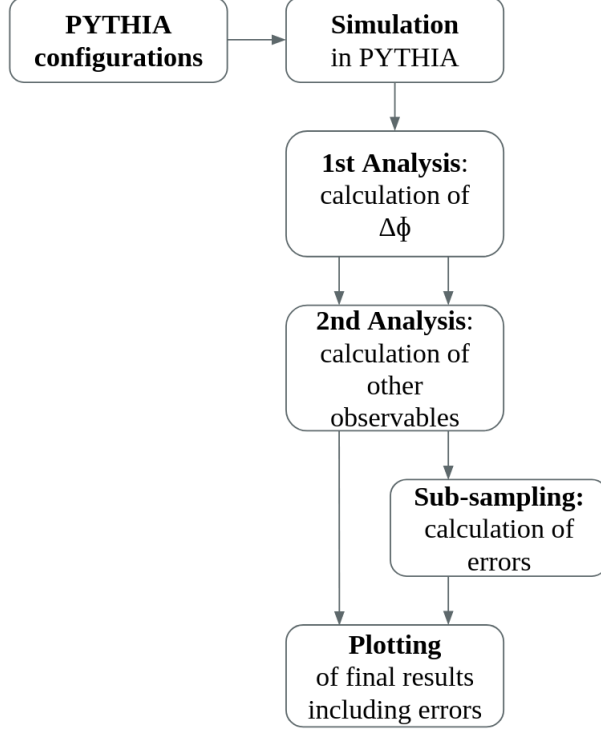


Figure 18: Flow chart indicating the full analysis procedure relevant to this thesis. Each of the stages is explained in more detail in the text. PYTHIA configurations refers to the input settings from which the simulations are started, and are given in appendix A.

The analysis is done using the PYTHIA event generator with the Monash and Junctions tunes and the ROOT data analysis framework. This section gives a systematic overview of the analysis steps and structure. Particular attention is given to the cuts applied in the simulation and analysis steps and the main background reduction method is described using LS and US correlations. The three stages of the analysis are: simulation, analysis and determination of statistical uncertainty, after which the results are obtained. The analysis flow is illustrated in Fig. 18 and the cuts are summarised in Tab. 1.

Simulation

The first stage of the analysis concerns the simulations done in PYTHIA of pp collisions at LHC energies of $\sqrt{s} = 14$ TeV. Around 1 billion events were simulated for both the Monash and Junctions tunes using hard $b\bar{b}$ (`HardQCD:hardbbbbar`) and hard $c\bar{c}$ (`HardQCD:hardccbar`) biased productions (`PYTHIA:QCDHardProcesses`) in order to increase statistics. The beauty and charm hadrons that are relevant for this study are explicitly kept from decaying in order to study their production

Table 1: Summary of cuts and model parameters in this analysis. The pseudorapidity cuts exactly match the coverage of the planned upgrade for the ALICE detector [12].

p_T [GeV]	> 0.15 (all particles) > 1.0 (trigger and associate)
$ \eta $	< 4
<code>ParticleDecays:tau0Max</code> [mm/c]	0.01 (10^{-14} seconds)
<code>PhaseSpace:pTHatMin</code> [GeV]	1

and two-particle angular correlations by setting a maximum decay time `ParticleDecays:tau0Max` (`PYTHIA:ParticleDecay`) on the average lifetime of the hadrons. This ensures that all particles within a given species are stable in the simulation. The limit is set to 0.01 mm/c, corresponding to 10^{-14} seconds, which is below the average lifetimes of all trigger and associate particles considered in the analysis. All particles are required to satisfy the conditions $p_T > 0.15$ GeV and $|\eta| < 4$ for the transverse momentum and pseudorapidity, respectively, and charged final-state particles are saved to measure event multiplicity. A technical summary of PYTHIA settings can be found in appendix A.

Analysis

In the first phase of the analysis the angular correlations are calculated from the PYTHIA event output and these correlations are divided into different categories associated with their event multiplicities. A trigger particle is chosen and associates are matched to this trigger, saving the azimuthal angle difference between these two particles and matched to the appropriate multiplicity bin. Note that both the trigger and associate particle are required to satisfy a $p_T > 1$ GeV cut in addition to the simulation cut. The charm sector benefits from a distinction into prompt and non-prompt production origin and therefore also divides the results accordingly. This is not done for the beauty hadrons as they can only be of prompt origin. The difference in prompt and non-prompt is addressed first using the mother particle(s) of the trigger and associate, as well as the angular correlations between their mothers, and the second part of the analyses focuses only on prompt (beauty and charm) contributions as this makes their comparison valid.

The second phase of the analysis is concerned with the calculation of the other observables. The angular correlation spectra are normalised by the appropriate number of triggers and the relevant trigger - associate pairs are plotted. Background is reduced by subtracting LS from US pairs. However, a note of caution is required here: the subtraction method will usually not always remove all background, as can be seen from the non-vanishing plateau after subtracting in Fig. 15. The effectiveness of the background estimation by the LS pairs is strongly correlated with the `PhaseSpace:pTHatMin` parameter (`PYTHIA:PhaseSpaceCuts`), which is set at a value of 1 GeV in this analysis. This dependence is sketched in Fig. 19 for US correlations and it can be shown that this parameter can be manipulated to obtain vanishing plateaus for the subtracted plots. The physical meaning of this parameter is obscure and it will therefore not be studied to further extent in this thesis.

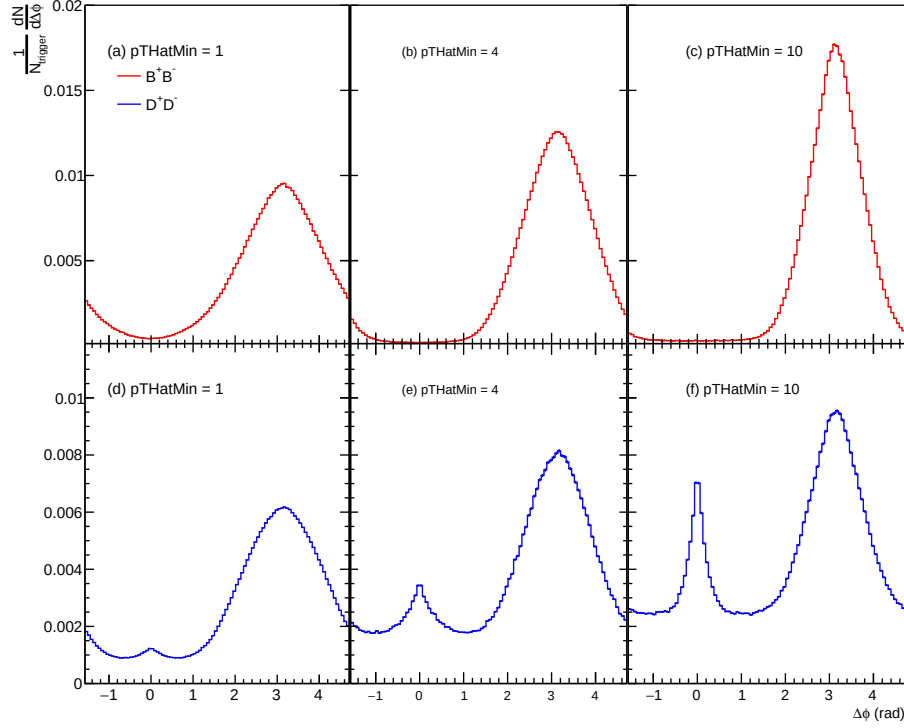


Figure 19: Angular correlations for (a) - (c) B^+B^- and (d) - (f) inclusive D^+D^- mesons over the 2π azimuthal range as simulated by PYTHIA with the Monash tune for different values of the p_{THatMin} parameter in units of GeV. The histograms are normalised by the total number of trigger B^+ or D^+ mesons, respectively. An increase in per-trigger yield is observed for increasing values of p_{THatMin} , as well as sharpening of the near-side peak in charm correlations.

Statistical uncertainty

A sub-sampling method is used in order to estimate the statistical errors on the balanced yield, baryon/meson ratio yield and Junctions/Monash ratios thereof, as they are strongly correlated. The PYTHIA output is divided into 20 equally-sized sub-samples, which are combined for the full analysis, but are kept divided for the sub-sampling procedure and passed to the second analysis phase. The resulting values are plotted and from the width of their distributions the standard deviations are estimated, which are interpreted as the error bars in the final result plots. An example is given in Fig. 20 for what these distributions look like (in the case of B^+B^-). The visualisation is strongly dependent on the binning and histogram range, but the idea is sketched and the correct error values, are indicated in the plot.

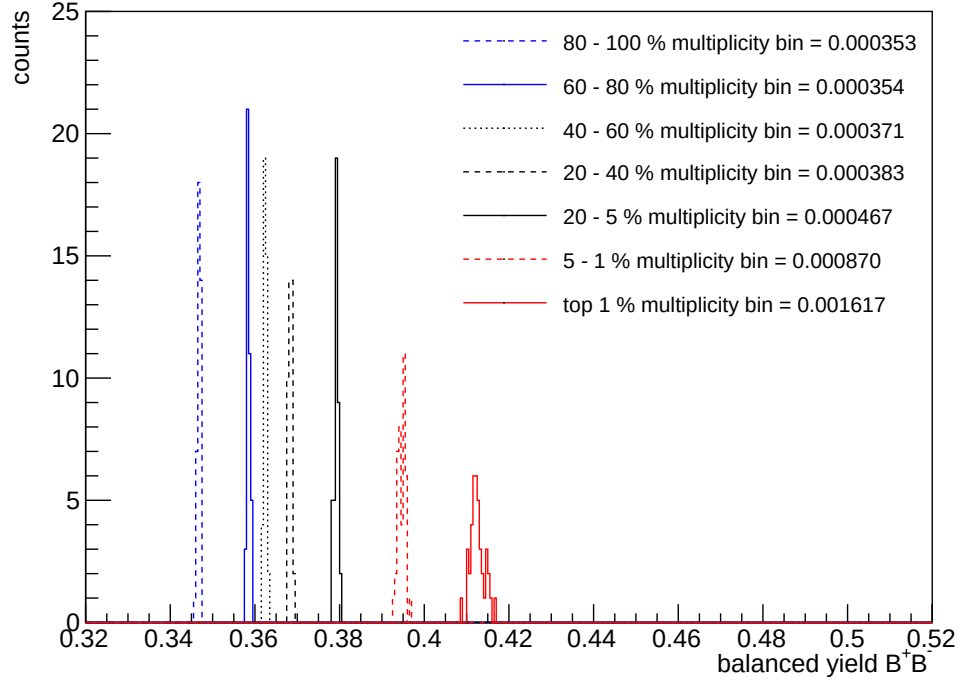


Figure 20: The result of sub-sampling the azimuthal integrated angular correlation yields for B^+B^- mesons is shown. The full PYTHIA Monash sample was divided into 40 equal sized samples and the results of the balanced yield is plotting in histograms for each of the multiplicity bins. The values given in the legend are estimated from a Gaussian fit by ROOT. Larger errors are seen in the sparse multiplicity bins.

4 Results

4.1 Angular correlations of B^+B^- and D^+D^- meson pairs

The heavy-flavour sector consist of beauty and charm hadrons which only differ in mass, hence their behaviour is expected to be similar. It is interesting to start the azimuthal angular correlations study by comparing D^+D^- and B^+B^- meson pairs so that any differences can be addressed and explained.

The differences in beauty and charm are most clear as a function of p_T and therefore the focus is shifted from multiplicity to p_T for this section. The regions are defined for the trigger-associate pairs as low-low ($p_T < 3$ GeV), intermediate-intermediate ($3 < p_T < 8$ GeV) and high-high ($p_T > 8$ GeV) and the number of pairs is normalised by the amount of trigger particles that satisfy the relevant p_T cut.

Fig. 21 shows the per trigger normalised yield for D^+D^- and B^+B^- pairs for different p_T trigger-associate regimes, where the positive-signed meson is taken as the trigger particle and the negative as associate. A defined peak is seen in the away-side region, as expected for heavy flavour mesons. The distinct difference between beauty and charm is observed in the near-side region, where only the charm mesons seem to be correlated. This behaviour hints at almost co-linear production of the D^+D^- pairs, which are not expected to originate from the initial scattering, and must have another origin. The remaining part of this section addresses this difference in beauty and charm angular correlations in the near-side region.

The nature of the produced D^+ trigger and D^- associate can be divided into two parts: (a) prompt production (hadronisation) or (b) non-prompt production (decay). If both trigger and associate are produced non-promptly and if they decayed from the same particle (the mother), it is expected that a near-side peak forms, as the decayed particles will be produced from their Lorentz boosted mothers co-linearly. Fig. 22 shows the angular correlations as in Fig. 21 but divided into the prompt and non-prompt production combinations that are possible for the trigger and associate particles. The figure clearly shows that the near-side peak is caused by indirect production mechanisms for both the D^+ and D^- mesons. Two questions still need to be addressed: 1. Which particles do the D^+ and D^- mesons decay from and 2. How often do the D^+ and D^- decay from the same particle?

The angular correlations in D^+D^- pairs in Fig. 23 are again divided into components, indicating the particles that the D^+ meson decayed from. The image is essentially identical for D^- with the appropriate changes for the charges (e.g. $D^*(2010)^+$ becomes $D^*(2010)^-$) and hence the discussion will only focus on the D^+ meson. The figure agrees with Fig. 22 showing that the near-side peak has a clear non-prompt origin. In addition, the figure also shows the abundances of the mothers from where it seen that most D^+ mesons originate from $D^*(2010)^+$ states, B^0 , $\bar{B}^0$, B^- or B^+ mesons.

If the D^+ and D^- decayed from the same mother then the near-side peak observed has a physical motivation. This can be shown by calculating the angular correlations of the mothers: about 13% of all D^+D^- pairs have the same mother, which are mostly B^0 , $\bar{B}^0$, B^- or B^+ mesons, while about 87% originates from the $D^*(2010)^{+/-}$ state. The latter is typically at the end of a decay chain pointing back to the same mother as the other correlated particle in the pair. For completeness, this is shown in appendix B.

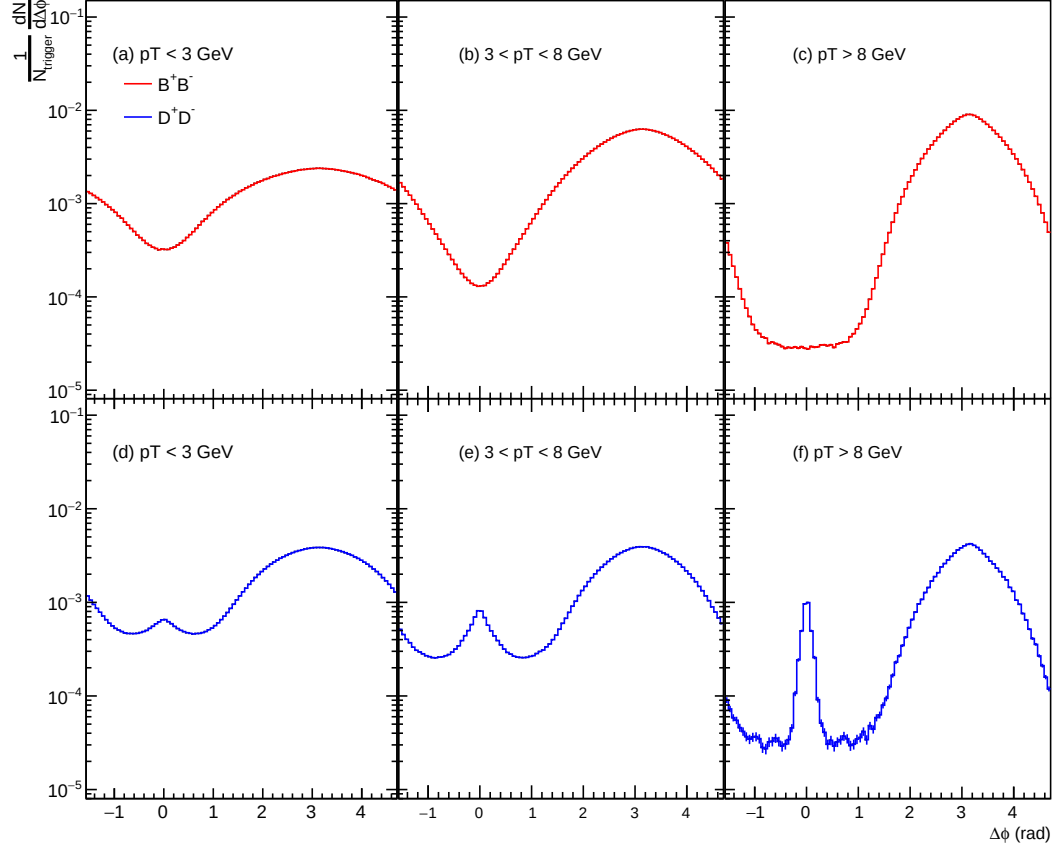


Figure 21: Angular correlations for B^+B^- (upper) and inclusive D^+D^- (lower) meson pairs in different p_T regions. A distinct away-side peak is observed for both meson pairs, while a near-side peak is only seen for the D^+D^- correlations.

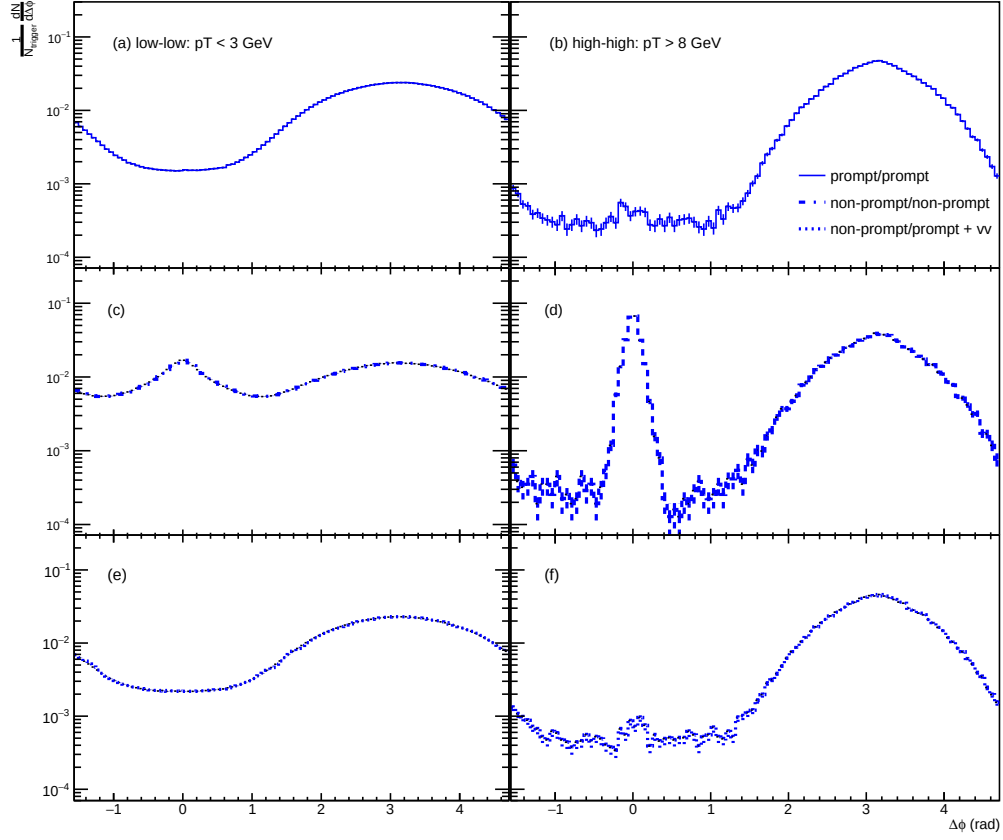


Figure 22: Angular correlations for D^+D^- meson pairs over the full multiplicity range and for low-low (left) and high-high (right) p_T regions. The correlations are divided into categories depending on the production origin of the D^+ and D^- mesons, which can be (a) - (b) prompt/prompt, (c) - (d) non-prompt/non-prompt, or (e) - (f) a combination of the two. Clearly, the near-side peak shows to be of a distinct non-prompt/non-prompt nature.

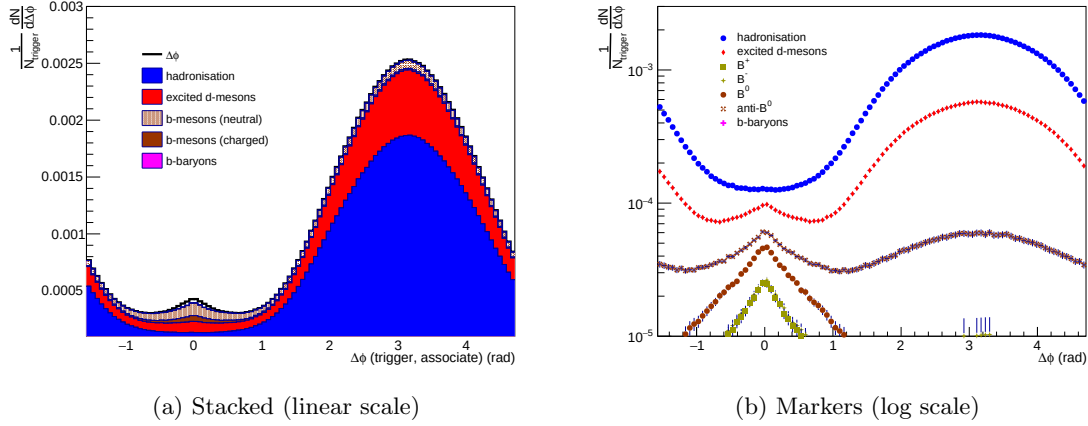


Figure 23: Angular correlations for D^+D^- meson pairs in the integrated p_T region. The correlations are divided into categories depending on the production origin of the D^+ meson, which can be prompt (hadronisation) or non-prompt (decay), the latter explicitly denoted by the mother of the D^+ meson. Both linear and logarithmic scales are used. From this figure it becomes clear that most D^+ mesons originate from $D^{*}(2010)^+$ states, B^0 , $\bar{B}^0$, B^- or B^+ mesons. The $\Delta\phi$ histogram is drawn as a consistency check, which is the sum of the individual components.

The comparison study of D^+D^- and B^+B^- meson pairs has shown that, even though both beauty and charm quarks are heavy-flavour and expected to behave similarly, there is still a pronounced difference in the near-side region of the azimuthal angular correlations. It is clear that there is a physics motivation for this behaviour arising from the fact that the D mesons can decay from the heavier B mesons, while the latter cannot decay from other mesons or baryons. The analysis is continued with prompt charm quarks in an attempt to equalise the comparison of the beauty and charm sectors.

4.2 Balancing of beauty and charm hadrons

The integrated yield plots are shown in Fig. 24 for B^+ and prompt D^+ meson trigger and appropriate associate particles as function of multiplicity. Focusing first on the beauty case, a distinct increase is seen in baryon balancing partners when switching on the Junctions tune compared with the Monash baseline. This increase from Monash to Junctions seems to be only weakly dependent on multiplicity, though the yield itself increases for larger multiplicity due to the increased number of MPIs, allowing the Junction mechanism to perform better as there will be a larger chance that different MPI states overlap in the same phase space. It should be noted that the increase in baryon yield for Junctions is especially strong for Σ_b baryon associates: an order of magnitude larger than the increase in the almost identical Λ_b^0 baryons. This is explained by the fact that the uu and dd di-quarks are necessary to create the $\Sigma_b^{+/-}$ baryons and can only exist in a isospin-1 state, while the Σ_b^0 baryon contains a ud di-quark that can reside in the much more energetically favoured isospin-0 state. This suppression is lifted when CR mechanisms are at play as they allow hadronisation without spin dependence. The Σ_b baryons are therefore created with equal probabilities and this is

reflected in Fig. 24. Both the Monash and Junctions tunes apply CR mechanisms, but the beyond leading colour treatment in the latter enhances this production in Σ_b baryons even more.

Turning to the charm sector, the trends seen in the beauty balancing results are mostly reproduced for charm. One of the differences is the substantial multiplicity dependence of the Junctions enhancement ratio for the $\overline{\Sigma}_c$ baryons, while the Σ_b baryons ratios were almost constant as a function of multiplicity. This seems to be related to how PYTHIA treats differences in Σ_b and $\overline{\Sigma}_c$ baryons as the effect is not seen for the Λ_b^0 and $\overline{\Lambda}_c^+$ baryons.

Another difference can be seen in the enhanced production of $\overline{\Lambda}_c^+$ baryons in Junctions, which is less pronounced than the equivalent beauty case. This again hints at differences in beauty and charm within PYTHIA and does not seem to be related only to mass effects. An explicit example of how this difference manifests is found in the `StringFlav:ProbQQ1toQQ0join` parameter ([PYTHIA:FlavourSelection](#)). This parameter controls the suppression of spin-1 di-quarks production relative to spin-0 when two quarks are combined to a di-quark in the Junctions framework. The default value (for Monash) when the heaviest quark is charm is 0.9 and 1.0 for beauty, giving an example of differences in beauty and charm implementation that can be argued to be causing the observed differences. The parameter has identical values in the Junctions tune, which is why this only explains differences in Monash. Some caution is required in the discussion: there seems to be some disagreement with data when the default setting is used and this is being tuned [44], however, and as there is no data available for the equivalent beauty case. Specifically, equalising the beauty and charm values for the `StringFlav:ProbQQ1toQQ0join` setting is not recommended and therefore not pursued.

4.3 Baryon/meson balancing ratio of beauty and charm hadrons

The baryon/meson balancing ratios are given in Fig. 25 for B^+ and prompt D^- trigger particles for different associate baryons as a function of multiplicity. The relative Λ_b^0 and $\overline{\Lambda}_c^+$ ratios are treated separately from the combined case including the Σ_b^0 and $\overline{\Sigma}_c^+$ baryons as the difference in yields differ up to an order of magnitude from each other.

A distinct increasing trend is observed in the baryon/meson ratio with increasing multiplicity in the Junctions tune for roughly all considered balancing partners. This is to be expected as the events with higher multiplicity are created through an enhanced number of MPIs, suspected to create more junction topologies in the hadronisation process, increasing the number of baryon partners. This trend is broken however in the higher multiplicity bins, where the ratio actually decreases. For beauty this occurs only in the highest multiplicity interval, but for charm this behaviour sets in already at mid-multiplicities in the 20-40 multiplicity bin due to the difference in mass.

Beside the differences in multiplicity dependence, there are clear differences between beauty and (prompt) charm, while naively it is expected that the prompt nature of the charm particles will cancel any differences with the beauty counterparts. The ratio seems to be particularly different among the considered particles for the Monash tune which hints at differences in the treatment of beauty and charm in PYTHIA instead of differences caused by the beyond leading colour mechanisms or originating from a physical origin. This belief is enhanced by the fact that the `StringFlav:ProbQQ1toQQ0join` parameter has different values for beauty and charm in Monash, but not in Junctions.

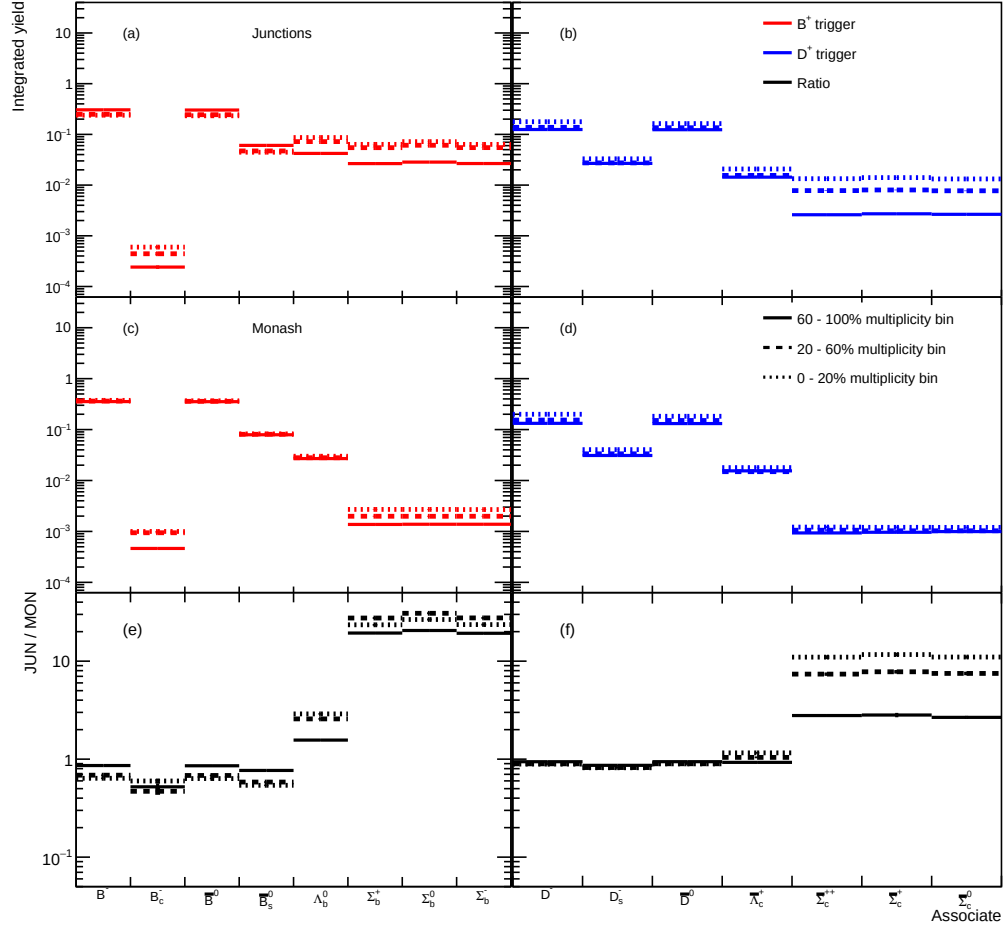


Figure 24: Integrated 2-particle angular spectra over the full 2π azimuthal range for (a), (c) B^+ and (b), (d) prompt D^+ mesons with respective beauty and prompt charm associates as a function of charged-particle event multiplicity. Both the PYTHIA Monash (lower) and Juncions (upper) tunes are used for the simulation of events. A clear increase in integrated yield is observed as a function of increasing multiplicity and an enhancement of baryon balancing partners is seen when enabling the Juncions tune in the simulation process due to the beyond leading colour CR mechanism.

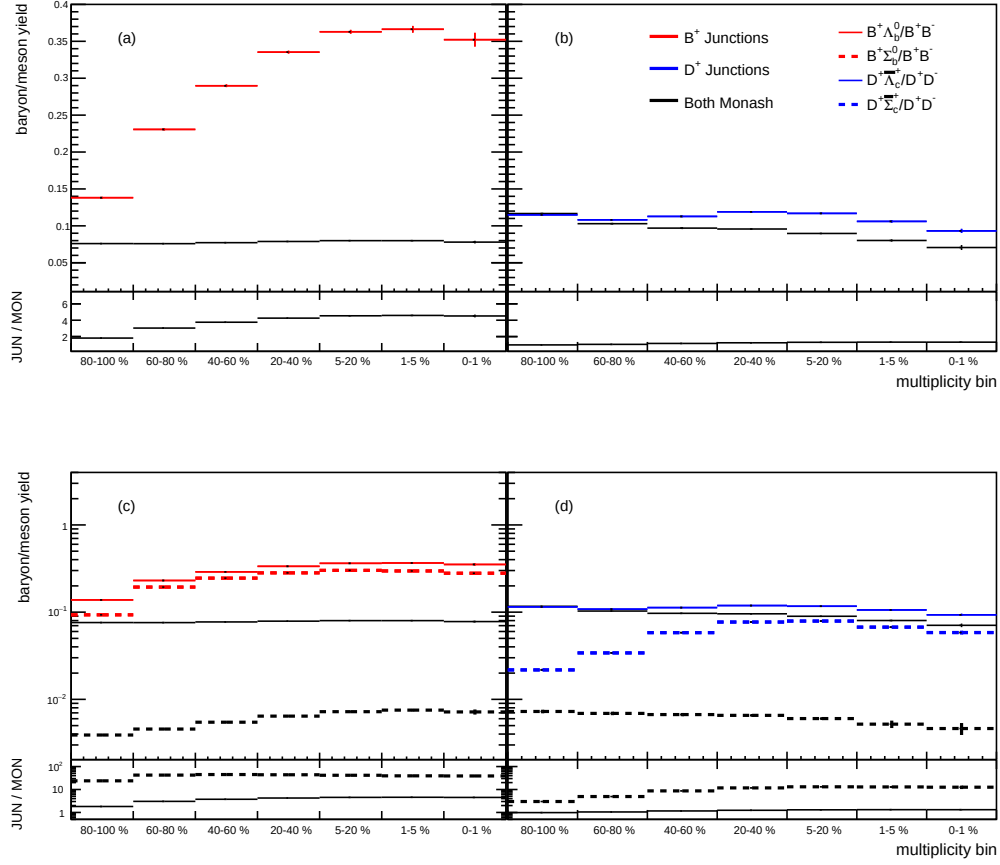


Figure 25: Baryon/meson ratios obtained from the integrated 2-particle yields in Fig. 24 with charged-particle event multiplicity dependence on the x-axis. The $B^+\Lambda_b^0/B^+B^-$ and $D^+\Lambda_c^+/D^+D^-$ pair ratios are shown in the upper plots (a), (b), while $B^+\Sigma_b^0/B^+B^-$ and $D^+\Sigma_c^+/D^+D^-$ are also included in the lower plots (c), (d) with a log-scale to visualise the increased yield in the latter. Both the PYTHIA Monash (black) and Junctions (red or blue) tunes are used and the charm particles are only produced promptly to provide a fair comparison with their beauty counterparts. Some similarities are seen in beauty and charm and the increase in baryon/meson ratio as a function of multiplicity is promising. The distinct saturation and decrease in baryon/meson ratio is, however, not expected intuitively and there seem to be differences in the treatment of beauty and charm that are not only caused by the mass difference in these sectors, in particular in the Monash tune.

5 Conclusion and Outlook

Hadronisation of beauty and charm quarks was discussed in this thesis using angular correlations and balancing as a function of event multiplicity. Simulations were done with the PYTHIA event generator in preparation for the increase in statistics and coverage of the upgrade to the current ALICE detector which will be necessary to study heavy-flavour correlations. The analysis commences with inclusive beauty and charm samples, but this shows a near-side peak in the D^+D^- angular correlations, argued to originate from non-prompt contributions in the charm sector. A similar scenario is not possible for beauty hadrons, as the top quarks are so heavy that they decay before they can hadronise. In order to have an appropriate comparison between charm and beauty, the analysis shifts to study prompt beauty and charm quarks.

Next, the analysis continues with the investigation of the balancing of B^+ and D^+ trigger mesons by several associate species of beauty and charm hadrons respectively. Background is reduced by subtracting LS pairs, which are a proxy for the full background of the event. The PYTHIA Junctions tune shows a clear increase in baryon associate yield in comparison to the Monash baseline as is expected from the new hadronisation topologies in Junctions that allow to directly create baryons. This yield is also enhanced as a function of multiplicity, expected because the event multiplicity increases as the number of MPIs increase, realising a larger probability of forming junction topologies. A noticeable larger enhancement is seen for the Σ_b and $\overline{\Sigma}_c$ baryons compared to the Λ_b and $\overline{\Sigma}_c$ baryons and in the case of the $\overline{\Sigma}_c^+$ this increase is strongly dependent on multiplicity. This enhancement is related to isospin suppression of the di-quark states that form the Λ (spin-0) and Σ (spin-1) baryons and linked to the `StringFlav:ProbQQ1toQQ0join` parameter in PYTHIA. Differences in beauty and charm are relatively minor and likely related to the treatment of these sectors in PYTHIA, specifically to the explicit differences in the default settings of the `StringFlav:ProbQQ1toQQ0join` setting that translate into differences between the beauty and charm baryons. The root of these differences stems from the many more measurements that have been done for charm with respect to beauty; in the latter most baryons have not even been observed yet.

Following the first balancing studies, the baryon-over-meson ratios are compared as a function of event multiplicity. The Junctions tune shows an increase in the ratio as the multiplicity grows due to the larger number of MPIs, indicative of better performance at balancing the trigger particles by hadrons. The trend stagnates and drops however for the events with the largest multiplicities, symbolising an effect which is not expected naively. This behaviour is qualitatively copied for all baryons considered in the Junctions tune. The Monash tune remains constant in most baryons, as expected, though shows a decreasing trend for the $D^+\overline{\Lambda}_c^+/D^+D^-$ ratio and the ratio of Junctions over Monash does not agree on any common behaviour between the particle species in both beauty and charm.

The differences in beauty and charm might arise from how these sectors are treated in PYTHIA. One of the ways in which this can be studied is by looking at the kinematics of the colour-singlet system with the invariant masses of the hadronisation products. Another approach to track the hadronisation at parton level in the event generator. This provides insight in how the generator evolves from the coloured beauty and charm quarks to their final state hadrons. Both of these studies could explain differences in the treatment of beauty and charm in PYTHIA, and they could also shed light on the observed peak structure, be it from physical or from MC-technical origin.

Although the ALICE3 upgrade is still about a decade away, it is important to have a clear picture of

the physics programme that should be pursued and how this can be achieved with the new detector. The massive increase in statistics and coverage, in combination with the increased luminosity of the LHC, create the perfect conditions for the analysis of heavy-flavoured mesons and baryons. The study of hard probes in the QGP becomes central to the physics programme, but also the hadronisation mechanisms of these heavy-flavour quarks, in particular related to baryons, get pushed to the front page. Taking this into account, it would be extremely valuable to investigate the angular correlations and balancing studies in the same way as has been done in this analysis on actual data and in heavy ion collisions, for which equivalent MC studies can be performed.

To conclude, the descriptions of hadronisation mechanisms and MPIs, besides many others, as well as the performance and improvement of PYTHIA and other event generators in this context, will be allowed to be studied in great detail and the ALICE3 upgrade will significantly enhance our understanding of the strong force and the initial stages of our Universe.

References

- [1] E. Hubble. A relation between distance and radial velocity among extra-galactic nebulae. *Proceedings of the National Academy of Sciences*, 15:168–173, 1929.
- [2] A. A. Penzias and R. W. Wilson. A measurement of excess antenna temperature at 4080 mc/s. *The Astrophysical Journal*, 142:419, 1965.
- [3] Edmund Bertschinger. Cosmological perturbation theory and structure formation, 2000.
- [4] J. Rafelski. Connecting QGP-heavy ion physics to the early universe. *Nuclear Physics B - Proceedings Supplements*, 243-244:155–162, 2013.
- [5] D. R. Matravers. Steven weinberg: cosmology. *General Relativity and Gravitation*, 41:1455–1458, 2008.
- [6] History of the Universe, 1991. <https://cds.cern.ch/record/39397>.
- [7] K. Aamodt et al. The ALICE experiment at the CERN LHC. *JINST*, 3:S08002, 2008.
- [8] ALICE Collaboration. The alice experiment – a journey through QCD, 2022.
- [9] D. J. Gross and F. Wilczek. Ultraviolet behavior of non-abelian gauge theories. *Physical Review Letters*, 30:1343–1346, 1973.
- [10] H. D. Politzer. Reliable perturbative results for strong interactions? *Physical Review Letters*, 30:1346–1349, 1973.
- [11] Christian Bierlich et al. A comprehensive guide to the physics and usage of PYTHIA 8.3, 2022.
- [12] ALICE Collaboration. Letter of intent for ALICE 3: A next-generation heavy-ion experiment at the LHC, 2022.
- [13] T. Gehrmann and B. Malaescu. Precision QCD physics at the LHC. *Annual Review of Nuclear and Particle Science*, 72:233–258, 2022.
- [14] Luca Rottoli. Precision QCD at the LHC: from the structure of the proton to all-order resummations, 2018.
- [15] John C. Collins, Davison E. Soper, and George Sterman. Factorization of hard processes in QCD, 2004.
- [16] S. et al. Acharya. Λ_c^+ production in pp and in p -pb collisions at $\sqrt{s_{NN}} = 5.02$ tev. *Phys. Rev. C*, 104:054905, Nov 2021.
- [17] P. Skands, S. Carrazza, and J. Rojo. Tuning PYTHIA 8.1: the Monash 2013 tune. *The European Physical Journal C*, 74(8), August 2014.
- [18] Jesper R. Christiansen and Peter Z. Skands. String formation beyond leading colour. *Journal of High Energy Physics*, 2015(8), August 2015.
- [19] V. Khachatryan et al. Evidence for collectivity in pp collisions at the lhc. *Physics Letters B*, 765:193–220, 2017.

- [20] Paul Langacker. *The standard model and beyond*. 2010.
- [21] Vernon D. Barger and R. J. N. Phillips. *Collider physics*. 1987.
- [22] J. F. Donoghue, E. Golowich, and Barry R. Holstein. *Dynamics of the standard model*, volume 2. CUP, 2014.
- [23] R. Keith Ellis, W. James Stirling, and B. R. Webber. *QCD and collider physics*, volume 8. Cambridge University Press, 2011.
- [24] Howard Georgi. *Lie algebras in particle physics*, volume 54. Perseus Books, Reading, MA, 2nd ed. edition, 1999.
- [25] Michael E. Peskin and Daniel V. Schroeder. *An Introduction to quantum field theory*. Addison-Wesley, Reading, USA, 1995.
- [26] Mark Thomson. *Modern particle physics*. Cambridge University Press, New York, 2013.
- [27] R. L. Workman et al. Review of Particle Physics. *PTEP*, 2022:083C01, 2022.
- [28] E. Norrbin and T. Sjöstrand. Production and hadronization of heavy quarks. *The European Physical Journal C*, 17(1):137–161, October 2000.
- [29] Michelangelo L. Mangano, Paolo Nason, and Giovanni Ridolfi. Heavy-quark correlations in hadron collisions at next-to-leading order. *Nuclear Physics B*, 373(2):295–345, 1992.
- [30] Matteo Cacciari, Mario Greco, and Paolo Nason. The pT spectrum in heavy-flavour hadroproduction. *Journal of High Energy Physics*, 1998(05):007–007, May 1998.
- [31] Zebo Tang, Nu Xu, Kai Zhou, and Pengfei Zhuang. Charmonium transverse momentum distribution in high energy nuclear collisions. *Journal of Physics G: Nuclear and Particle Physics*, 41, 12 2014.
- [32] Rainer Sommer. *Non-perturbative renormalization of QCD*, page 65–113. Springer Berlin Heidelberg.
- [33] Gunnar S. Bali, Hartmut Neff, Thomas Düssel, Thomas Lippert, and Klaus Schilling. Observation of string breaking in QCD. *Physical Review D*, 71(11), June 2005.
- [34] G.S. Bali, T. Düssel, T. Lippert, H. Neff, Z. Prkacin, and K. Schilling. String breaking. *Nuclear Physics B - Proceedings Supplements*, 153(1):9–16, March 2006.
- [35] B. Andersson, G. Gustafson, G. Ingelman, and T. Sjöstrand. Parton fragmentation and string dynamics. *Physics Reports*, 97(2):31–145, 1983.
- [36] Bo Andersson, Sandipan Mohanty, and Fredrik Soderberg. Recent developments in the lund model, 2002.
- [37] Stefan Höche. Introduction to parton-shower event generators, 2015.
- [38] Torbjörn Sjöstrand and Maria van Zijl. A multiple-interaction model for the event structure in hadron collisions. *Phys. Rev. D*, 36:2019–2041, Oct 1987.

- [39] V. et al. Khachatryan. Charged particle multiplicities in pp interactions at $\sqrt{s} = 0.9$, 2.36, and 7 tev. *Journal of High Energy Physics*, 2011(1), January 2011.
- [40] G et al. Aad. Charged-particle multiplicities in pp interactions measured with the ATLAS detector at the LHC. *New Journal of Physics*, 13(5):053033, May 2011.
- [41] B. et al. Abelev. Multiplicity dependence of the average transverse momentum in pp, p–pb, and pb–pb collisions at the LHC. *Physics Letters B*, 727(4–5):371–380, December 2013.
- [42] Jesper R. Christiansen. Colour reconnection - models and tests, 2015.
- [43] Bo Andersson. *The Lund Model*, volume 7 of *Cambridge Monographs on Particle Physics, Nuclear Physics and Cosmology*. Cambridge University Press, 7 2023.
- [44] J. Altmann, A. Dubla, V. Greco, A. Rossi, and P. Skands. Towards the understanding of heavy quarks hadronization: from leptonic to heavy-ion collisions, 2024.

A PYTHIA simulation settings

The code below is used to set the PYTHIA simulation settings in this analysis. Several parts can be commented out or altered to differentiate in beauty, prompt charm or inclusive charm for both the Monash and Junctions tunes.

```
1 ! Excerpt of pythiasettings_Hard_Low_cc_JUNCTIONS.cmd
2 ! Called from file: ccbarcorrelations_status_JUNCTIONS.cpp
3 ! This script defines the settings in order to simulate collision events which are
4 !   used to calculate azimuthal angular correlations between charm hadrons.
5 ! Lines starting with (!) are comments
6
7 !!! General settings !!!
8
9
10 ! More information about these commands can be found on https://pythia.org/latest-manual/MainProgramSettings.html
11 Init:showMultipartonInteractions = off
12
13 ! Setting up beams and tune
14 ! More information about tunes can be found here: https://pythia.org/latest-manual/Welcome.html
15 ! More information about Beams parameters can be found here: https://pythia.org/latest-manual/BeamParameters.html
16
17 Beams:eCM = 14000          ! sqrt(s) = 14 TeV (eCM counts as GeV)
18 Beams:idA = 2212          ! beam 1 proton
19 Beams:idB = 2212          ! beam 2 proton
20 Tune:pp = 14              ! tune 14 = MONASH
21
22
23 !!! JUNCTIONS tune !!!
24 ! mode 2 QCD from Skands et al.
25 ! This part is used for Junctions productions, comment out for Monash.
26
27
28 StringPT:sigma = 0.335
29 StringZ:aLund = 0.36
30 StringZ:bLund = 0.56
31 StringFlav:probQQtoQ = 0.078
32 StringFlav:ProbStoUD = 0.2
33 StringFlav:probQQ1toQQ0join = 0.0275,0.0275,0.0275,0.0275
34
35 MultiPartonInteractions:pT0Ref = 2.15
36
37 BeamRemnants:remnantMode = 1
38 BeamRemnants:saturation = 5
39
40 ColourReconnection:mode = 1      ! 1 = More QCD scheme
41 ColourReconnection:allowDoubleJunRem = off
42 ColourReconnection:m0 = 0.3
43 ColourReconnection:allowJunctions = on
44 ColourReconnection:junctionCorrection = 1.20
45 ColourReconnection:timeDilationMode = 2
46 ColourReconnection:timeDilationPar = 0.1
47
48 ! End of JUNCTIONS settings
```



```

49
50
51 !!! Physics part !!!
52
53
54 ! Processes
55 ! Hard QCD processes: https://pythia.org/latest-manual/QCDHardProcesses.html
56
57 ! HardQCD:hardccbar is turned on for prompt charm production
58 ! HardQCD:hardbbbbar is turned on for prompt beauty production
59 ! HardQCD:hardccbar and HardQCD:hardbbbbar are both turned on for inclusive charm
    production
60
61 ! HardQCD:all = on
62 HardQCD:hardccbar = on
63 ! HardQCD:hardbbbbar = on
64
65 ! Processes become divergent at low pT, need to define a lower bound
66 PhaseSpace:pTHatMin = 1.
67
68 ! Decay supression
69 ! More information about decay supression can be found here: https://pythia.org/
    latest-manual/ParticleDecays.html
70
71 ParticleDecays:limitTau0 = on      ! Particles with tau0 > tau0Max are not allowed to
    decay.
72 ParticleDecays:tau0Max = 0.01      ! tau0Max is suppressing the weak decays so the
    produced charm particles will not decay
73      ! This tau0max is expressed in mm/c which is around 1e-12 seconds

```

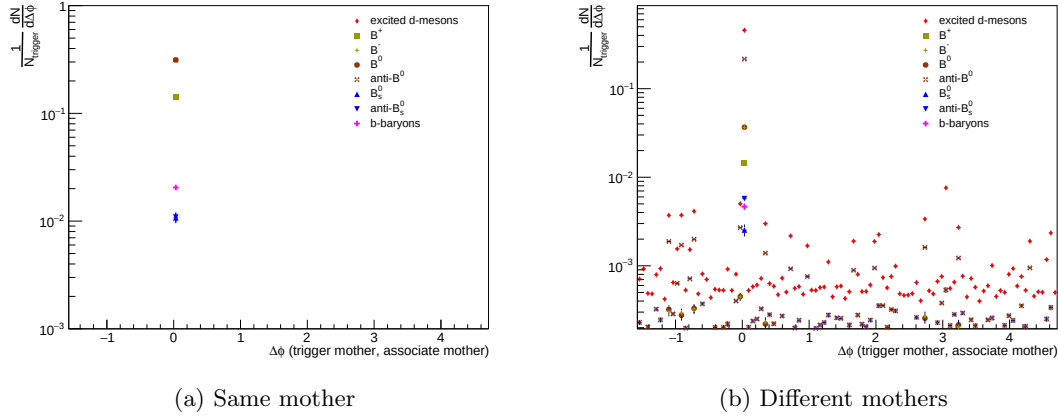


Figure 26: Per trigger normalised yield histograms of azimuthal angular correlations of the mothers of the D^+D^- meson pairs for the integrated p_T regions are shown. The correlations are divided into categories depending on the mother of the D^+ and D^- , which can be the same mother, or two different mothers. By inspecting the number of entries in the histograms (not presented) it can be shown that about 13% of all D^+D^- pairs have the same mother, which are mostly B^0 , $\bar{B}^0$, B^- or B^+ , while about 87% originates from the $D^*(2010)^{+/-}$.

B D^+D^- angular correlations of mothers

Fig. 26 shows the angular correlations of the D^+ and D^- mothers in the case when both originate from the same particle and when they have different mothers. The angular correlations of the mothers are shown for self-consistency, though it can be seen from those results that about 13% of all D^+D^- pairs have the same mother, which are mostly B^0 , $\bar{B}^0$, B^- or B^+ mesons, while about 87% originates from the $D^*(2010)^{+/-}$. In the case of direct decay from the same mother, the near-side peak is expected. In the case with different mothers, it should be noted that the $D^*(2010)^{+/-}$ state is the direct mother, but it typically forms a decay chain allowing other excited D states and B mesons to be further back, still pointing back to the same mother, which is the expected behaviour for a near-side peak forming in the 87% of cases with different direct mothers.